ÉCOLE DE TECHNOLOGIE SUPÉRIEURE
Université du Québec

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Rapport de projet Phase 3
LOG430 — Architecture logicielle
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École de technologie supérieure

Run book

Prerequisites

To run this project locally, you need the following tools installed:

- Go 1.21+
- Docker & Docker Compose

Installation

1. **Clone the repository**

```
git clone https://github.com/cjayneb/log430-project.git
cd log430-project
```

2. **Configure environment variables**

   Change the values in the `backend/services/{service-name}/config.go` files if you want to override defaults. Example:

```
 Port  string `env:"APP_PORT" envDefault:"8181"`
 DBUrl string `env:"DATABASE_URL"
envDefault:"user:pass@tcp(127.0.0.1:3306)/brokerx"`
```

> The variables in `config.go` are used when running the app locally. The environnement variables set in the `docker-compose.yml` will override the values set in `backend/services/{service-name}/config.go` when running the project with Docker Compose.

Running the project

Run locally (without Docker)

From on of the `backend/services/{service-name}` directories:

```
go run .
```

This will start the microservice's API on http://127.0.0.1:8080.

- Health endpoint: http://127.0.0.1:8080/health (GET) (each microservice has one)

More examples

- Login endpoint: http://127.0.0.1:8080/api/user/auth/login (POST)
- Home page endpoint: http://127.0.0.1:8080/ (GET)
- Orders page endpoint: http://127.0.0.1:8080/api/order (GET)

> You must have a MySQL instance and a Redis instance running on your machine for all uses cases to work

Run with Docker Compose

From the project root:

```
docker compose down -v # To remove existing containers and their volumes
docker compose up --build -d
```

This starts:

- The Go backend microservices on http://localhost/api
- A MySQL database (`brokerx_db`) on port `3306`
- Nginx API Gateway
- Redis DB

Running tests

Important : Before running all tests, you must have the test database up, because some of the tests are integration tests needing a real database.

Run the following command to start the test database:

```
#From the root of the project
docker compose -f docker-compose.test.yml up -d
```

Running all tests (with coverage report)

Inside the `backend/services/{service-name}` folder of each micro-service:

```
go test ./... -coverprofile=coverage
```

Generate HTML coverage report

```
go tool cover -html=coverage
```

Deployment

At this stage, the application is deployed locally or remotely using Docker Compose. A production-ready deployment would likely use Kubernetes or cloud-based services, but that is outside the current scope.

Deploying locally

To deploy locally, you just have to run the following commands

```
docker compose down -v # Ensure Docker is clean with no previous
deployment
docker compose up --build -d # Build the Docker image and run docker
compose in detached mode
```

Deploying remotely

The GitHub Actions Workflow should take care of deploying the application to the ETS Virtual Machine self hosted runner automatically on every push. See `.github/workflows/ci_cd.yml`

The current deployment pipeline is broken because of issues with the storage on the ETS VM.

> To access the remote deployment, you must be connected to the ETS Cisco Secure Client via accesvpn.etsmtl.ca

User Guide

This document show syou how to use BrokerX once it has been deployed either locally or remotely.

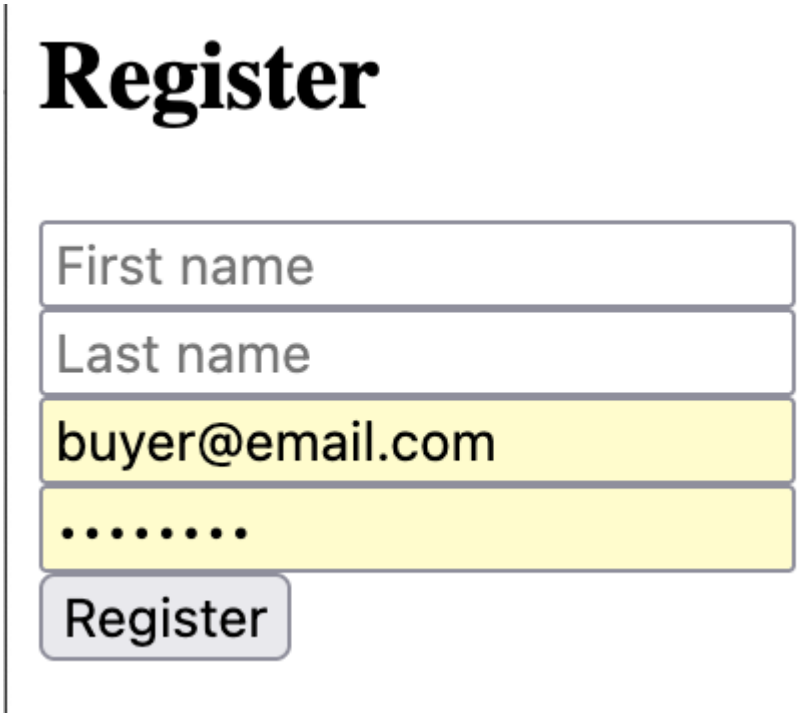
Access BrokerX

To access BrokerX, use your web browser and navigate to :

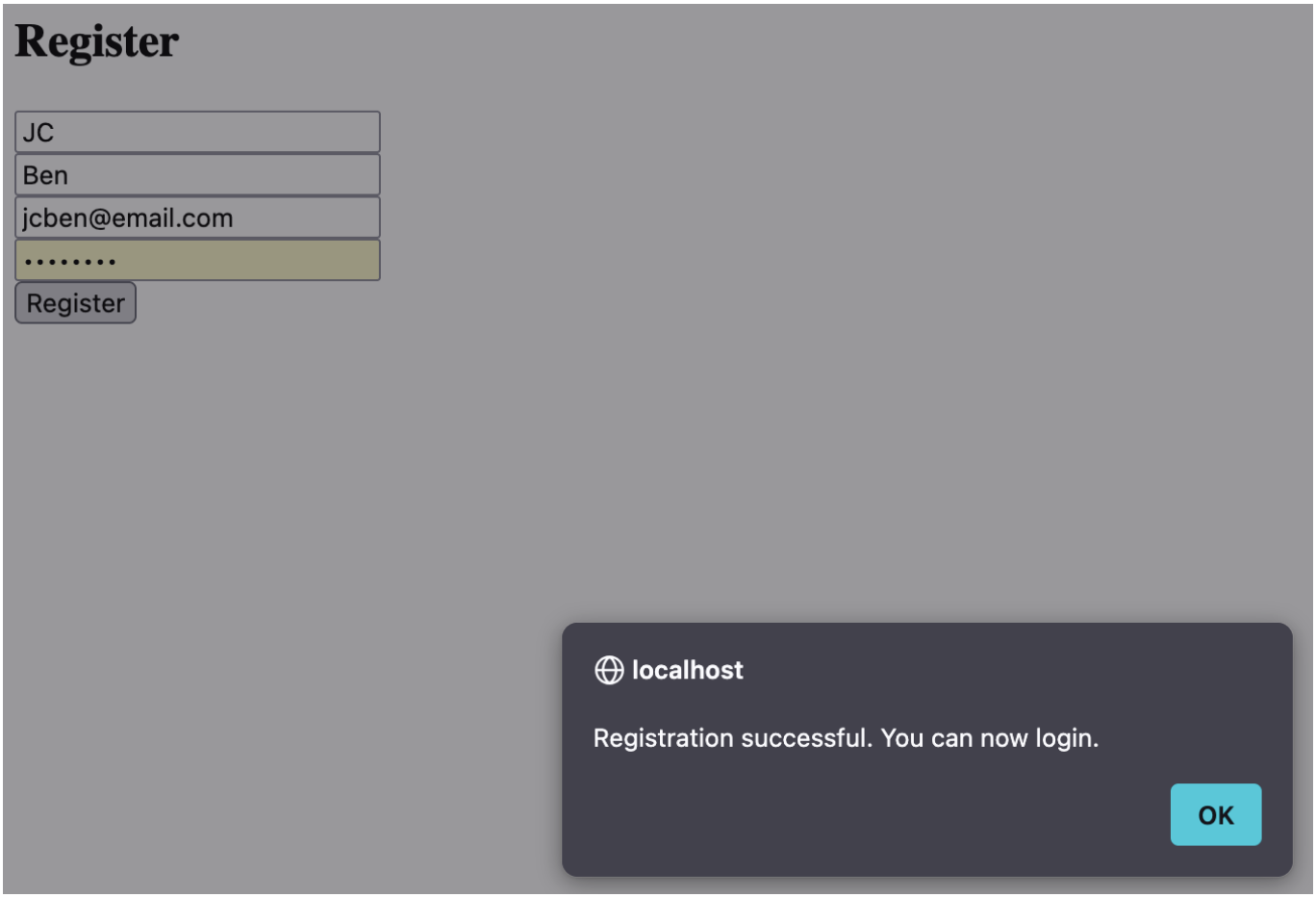
- Locally : http://localhost/
- Remotely : http://10.194.32.206/

Registering as a new user

Access the following link in your browser : http://localhost/register.html



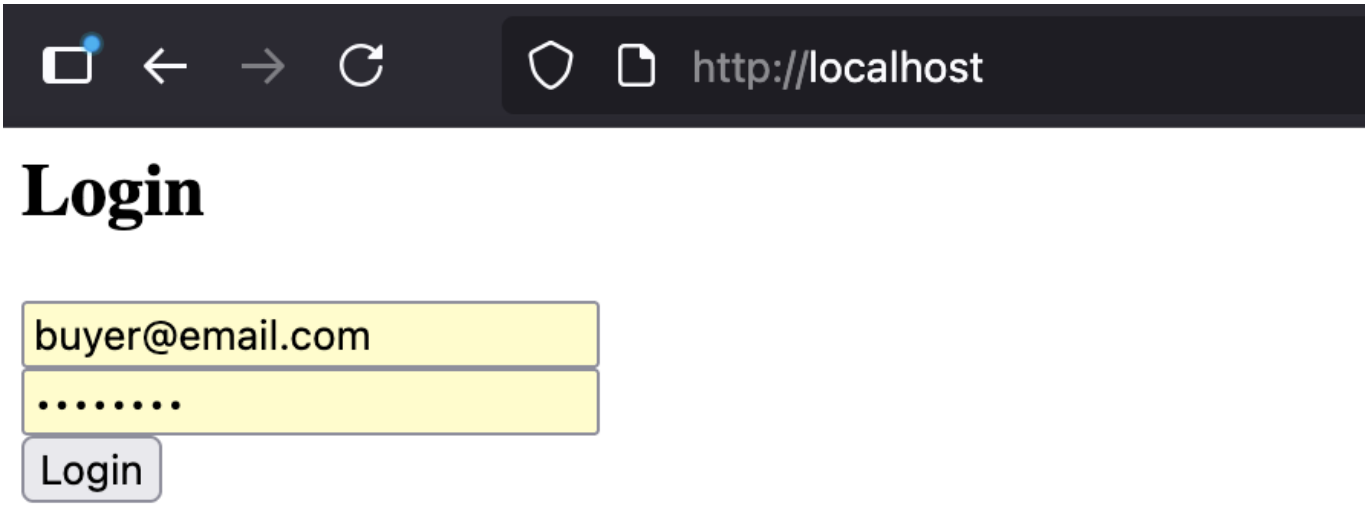
Enter you first and last name as well as an email and a password and submit :



Logging in

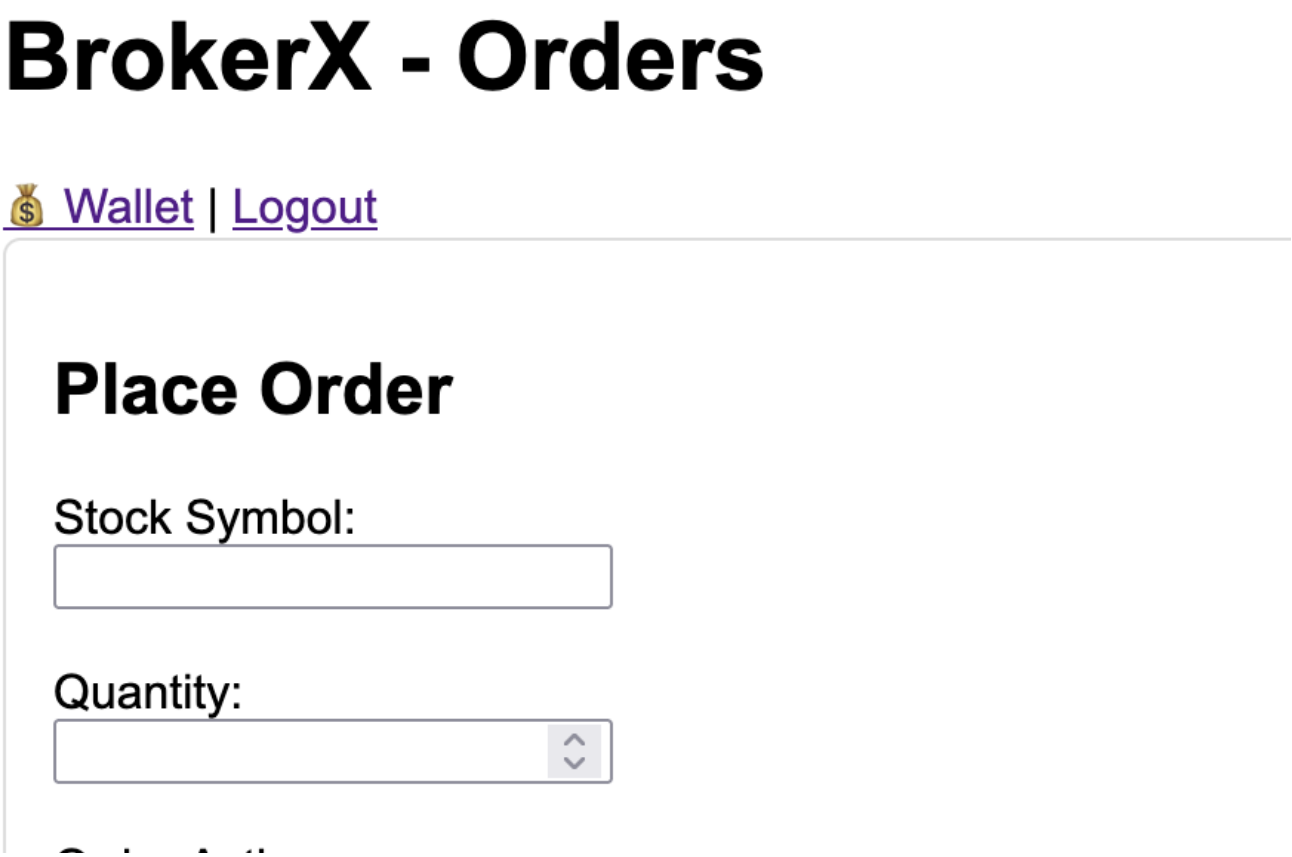
Enter the email and password and click on *Login* to access the account of a seller :

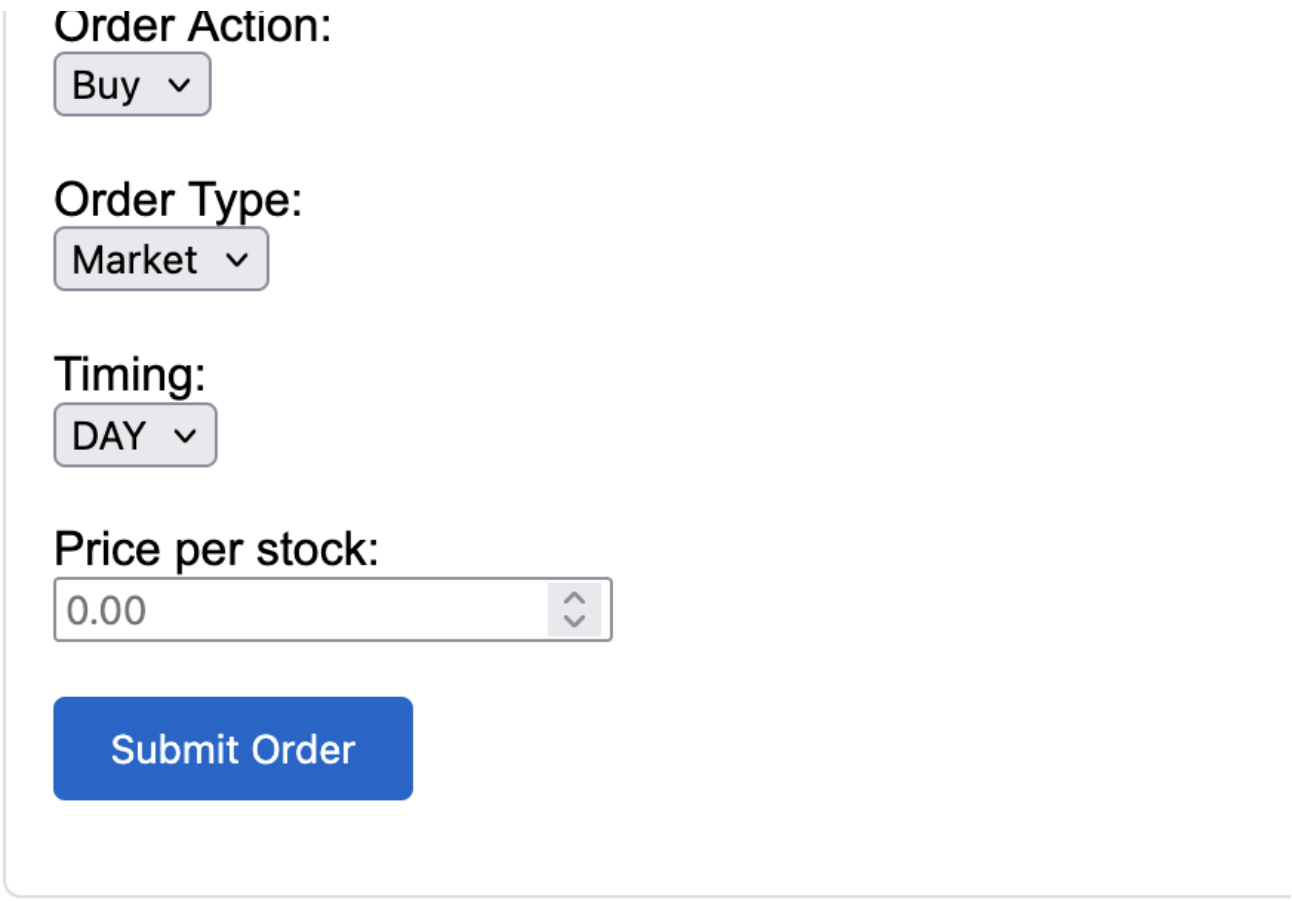
- Email : seller@email.com
- Password : password



Placing an order

Logging in will bring you to this *Orders* page:

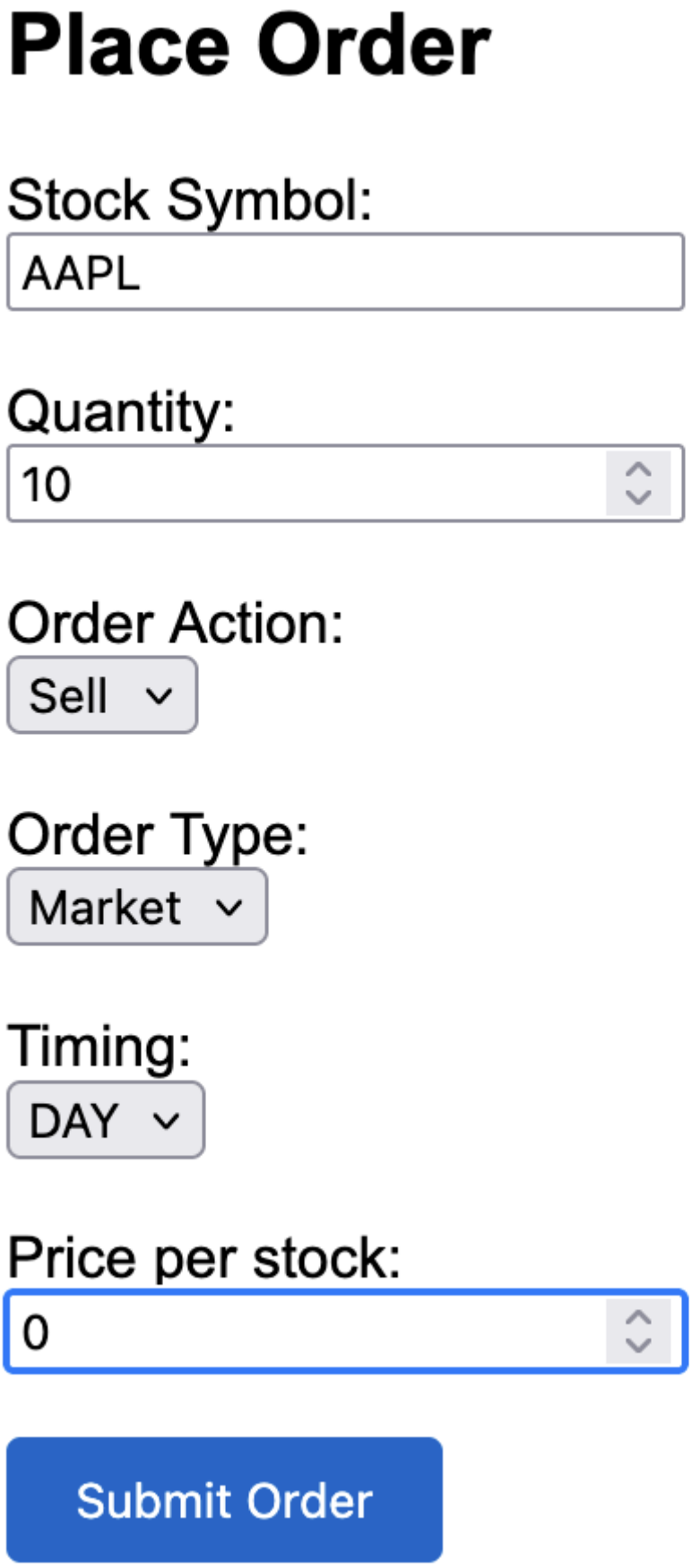




Your Orders

No orders yet.

Then you can fill the form to sell AAPL stocks at market price and click *Submit Order* :



Then you should see your order appear at the top of the page and this confirmation message at the bottom of the form.

> Error messages will also appear at the bottom of the form.

Submit Order

Order placed successfully.

Your Orders

ID	Symbol	Quantity	Remaining	Action	Type	Timing	Unit Price	Status	Created At
1	AAPL	10	10	sell	market	day	0	open	2025-10-29, 12:06:03 a.m.

Next, if you refresh the page, you should see that your order status, and remaining quantity have been updated if a matching order has been found :

Submit Order

Your Orders

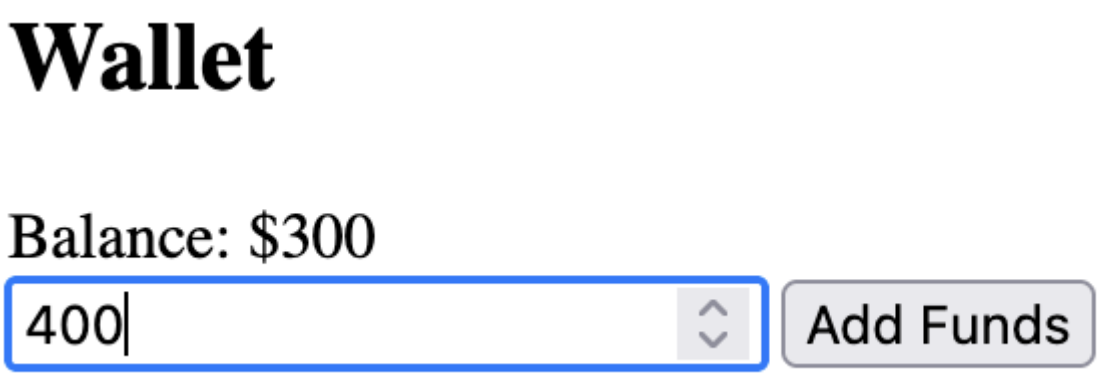
ID	Symbol	Quantity	Remaining	Action	Type	Timing	Unit Price	Status	Created At
1	AAPL	10	4	sell	market	day	0	partially_filled	2025-10-29, 12:06:03 a.m.

> Other users (email and buyer@email.com) have different base positions, orders and wallet balances, which make for different outputs in results.

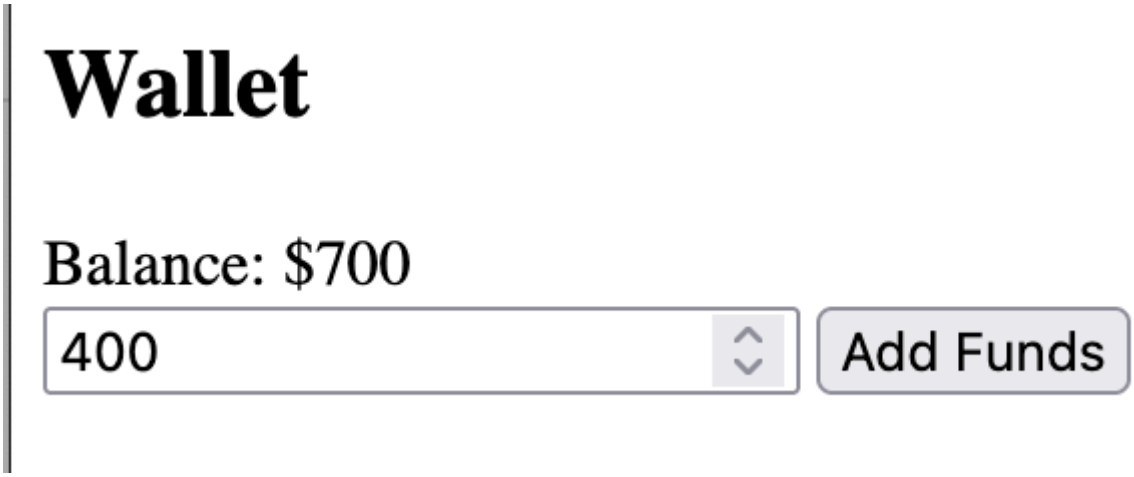
Adding funds to your wallet

After logging in, it is possible to add funds to your wallet by clicking on *Wallet*:

Then entering the amount you want to deposit and submitting :



The balance automatically updates itself and you will be able to make bigger purchases of stocks :



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1. Introduction and Goals

1.1 Requirements Overview

The BrokerX project is an online brokerage platform targeting individual investors, developed in response to the rapidly evolving financial services sector. It is made to allow retail investors to trade stocks securely and rapidly from the comfort of their home.

The system **must** support the following core functionalities:

- **Authentication** : Secure login with session cookies.
  - Justification: Secure access to the system is mandatory, especially considering the sensitivity of the data residing on BrokerX.
- **Order placement** : Allow users to place market or limit orders on various stocks.
  - Justification: This is a core business function. Without order placement, Brokerx has no reason to exist.
- **Matching, Execution & Confirmation** : Ensure matching of buy and sell orders using an internal matching engine, generating an execution report and balancing wallet and position quantities.

    - Justification: This is another feature that is too important to be ignored, placing an order to buy stocks is a must, but it is not useful if the order cannot be matched with a corresponding selling order.

The system **should** support the following functionalities:

- **Market data subscription** : Realtime market data updates (top-of-book, trades, OHLC).
    - Justifiifcation: This feature should definitely be a part of BrokerX, so that users do not have to use another platform to consult stock prices, but it is not a necessity to have a brokerage system that works.
- **Registration** : Account creation and identity verification via an OTP sent by email.
    - Justification: For now, user accounts can be manually added to the system and tests of the whole system can still be done without a registration page.
- **Wallet funding** : Allow users to add money to their BrokerX wallet in order to buy stocks and get compensated for selling stocks.
    - Justification: For now, wallets can be also be manually created and given any amount as a test balance, since the system does not yet interact with the real stock market.
- **Order modification/cancellation** : Allow users to modify or cancel order that have been placed.
    - Justification: This is not a necessity as it does not pose any real risk to lack the ability to modify or cancel orders during this phase. However, a brokerage platform should definitely have this feature in order to rectify input errors.

The system **could** support the following functionalities:

- **Notifications** : Send email notifications to the user when an event happens for a given order.
    - Justification: This is a nice-to-have and in no way necessary to a working brokerage platform.a

> These functions cover the core trading workflow and will be expanded in later iterations.

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1.2 Quality Goals

Priority	Quality goal	Scenario
1	Latency	≤ 100 ms to get an acknowledgement after placing an order.
2	Throughput	≥ 1000 orders successfully placed per second.
3	Availability	The system must be available at least 99.9% of the time.

These goals are to be met during the third iteration (event-driven architecture) of BrokerX.

Notes / roadmap: The project specification defines phased targets (monolith → microservices → event-driven) where the latency/throughput and availability targets tighten in later phases. Observability (logs + metrics) is required from phase 2 and distributed tracing from phase 3. These constraints drive the choice of architecture and incremental migration plan

1.3 Stakeholders

Contents.

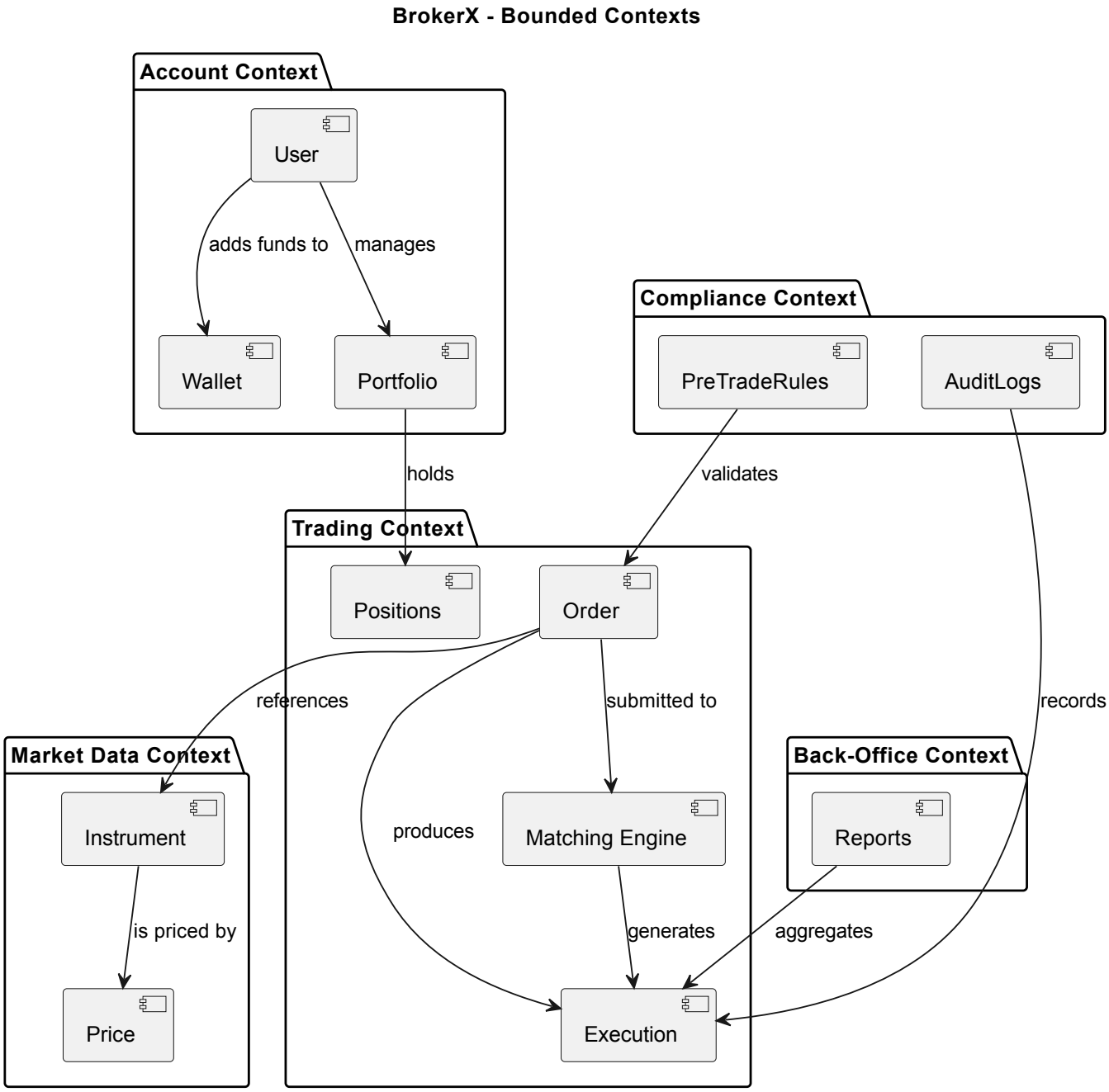
Role/Name	Contact	Expectations
Clients (web users)	*N/A*	Fast, correct order handling; clear execution confirmations; accurate portfolio balances.
Developers of BrokerX	Jean-Christophe Benoit	Clear modular code structure (domain vs infra), testability, reproducible CI/CD, containerized deployment.
Back-Office Operations	*N/A*	Accurate execution records, trustworthy reports for accounting.
Compliance /Risk Employees	*N/A*	Control over pre-trade checks, post-trade surveillance, immutable logs, idempotency guarantees.
External Market Data Provider	*N/A*	Well-defined streaming API (subscribe/unsubscribe); order routing contracts for simulated exchanges.

2. Architecture Constraints

Constraint	Description
Technology	C#, Go, Rus or C++ permitted. Python or JavaScript/TypeScript are strongly discouraged for the backend part of the system.
Performance	The system must meet latency, throughput and availability targets.
Deployment	The system prototype must be containerized and deployed on a public or semi publicly accessible platform via an automatic CI/CD pipeline. Multiple artifacts must be deployed during phase 2. The deployed instances must be horizontally scalable (stateless services).

3. System Scope and Context

Business Context



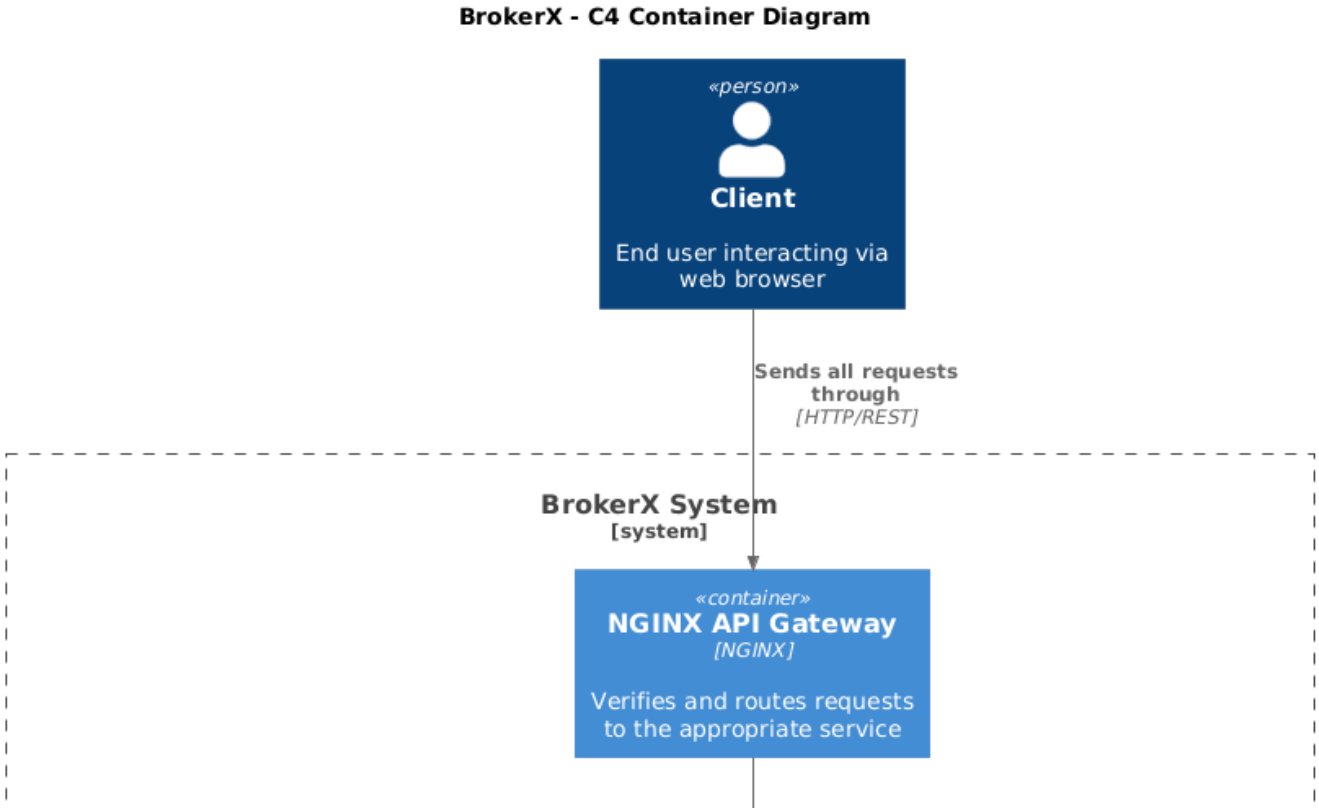
Technical Context

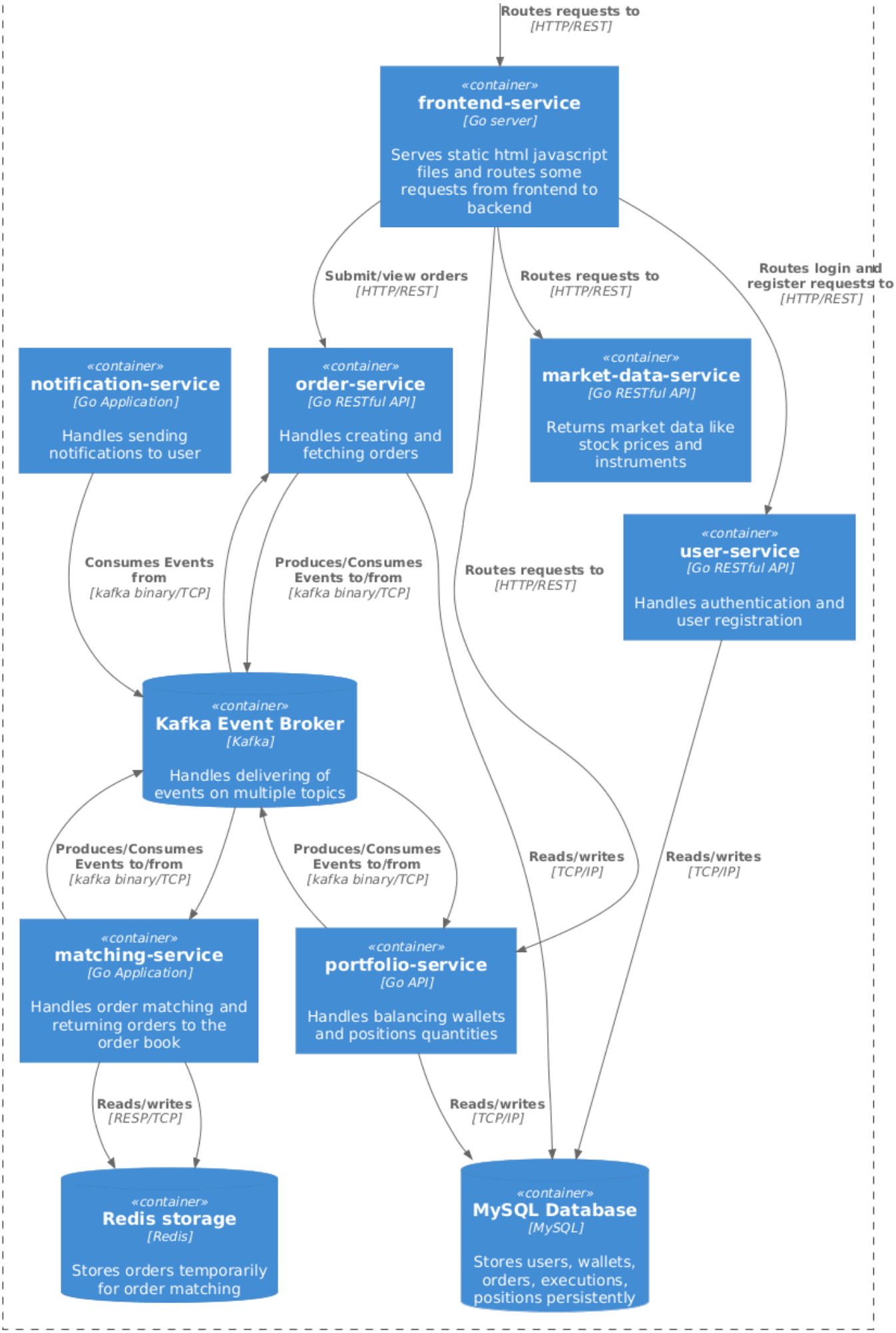
4. Solution Strategy

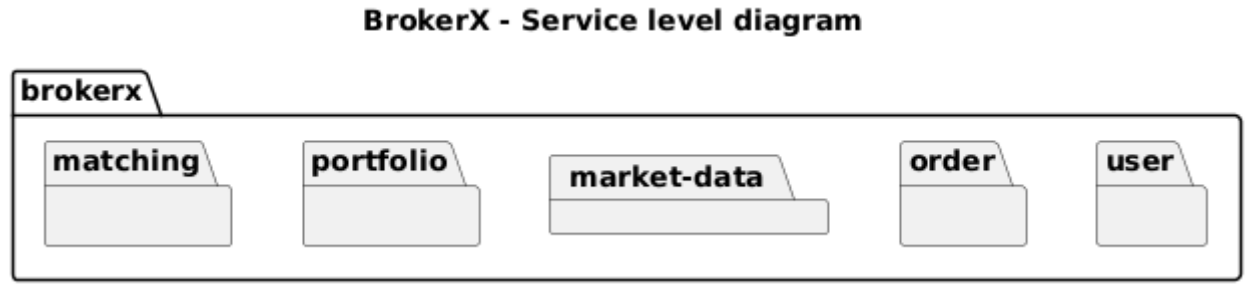
Problem	Solution
Maintainability and evolution	Internally, the system code is organized in a modular fashion using an architecture similar to hexagonal. This isolates the domain logic from infrastructure adapters. This allows the monolithic prototype to evolve somewhat easily in later phases.

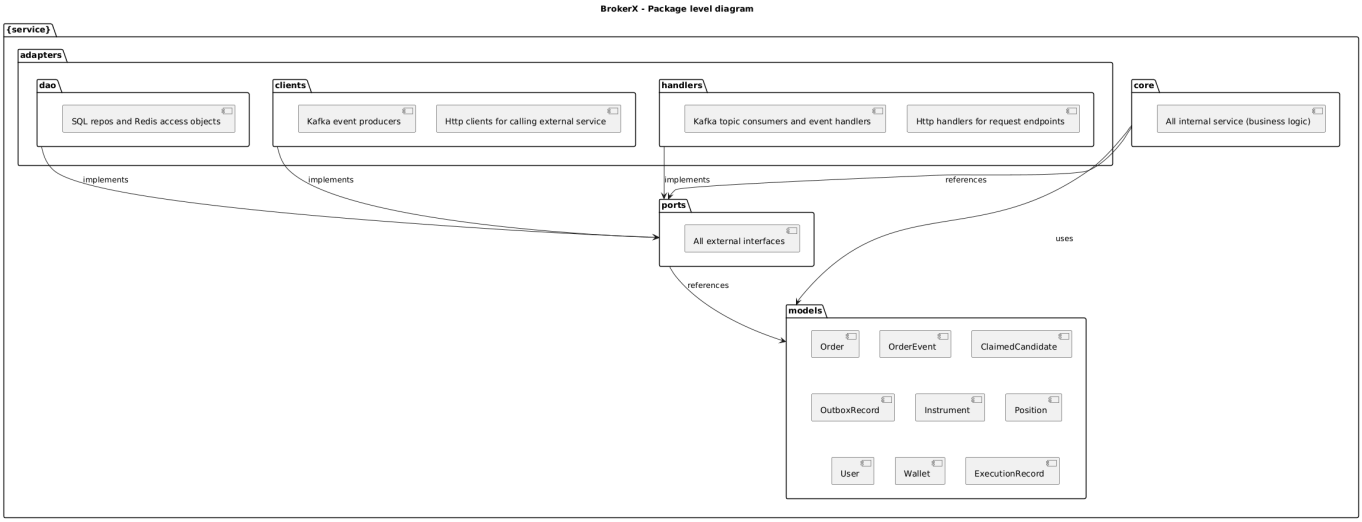
Problem	Solution
Data consistency and integrity	A MySQL database is used to store all data and acts as a persistent ledger. Entities and data queries and commands are managed with repositories which abstract the database specific logic into interfaces. Transactions are used in critical flows like order executions to maintain consistency. A Redis sorted set acts as a live order book. Kafka topics are used to send and consume events for each service. The system use the choreographed Saga pattern along with Outbox to prevent data losses in case of a crash.
Delivering data to client	The Go server supports multiple http endpoints for server rendered html templates and data retrieval. The frontend code can query these endpoints freely to get the desired data.
Latency and throughput	Using the Go language because it is well-suited for high concurrency, low latency environments. Goroutines are easily implementable for asynchronous processing like the outbox event dispatcher or the event consumers. Kafka is used to produce events, such as the OrderCreated event which is created as soon as an order is placed, making order acknowledgement faster.
Error Handling & Observability	The Go programming language is made for functions to return multiple return values, making it very easy to propagate errors from any layers back to the client. It also comes with a an integrated logging library, ensuring the system's behaviors and faults are observable at any point. Using the slog library along with the Go Context, a TraceID and the exact logging line can be found in each log, making it easy to know which request had which effect on the system.

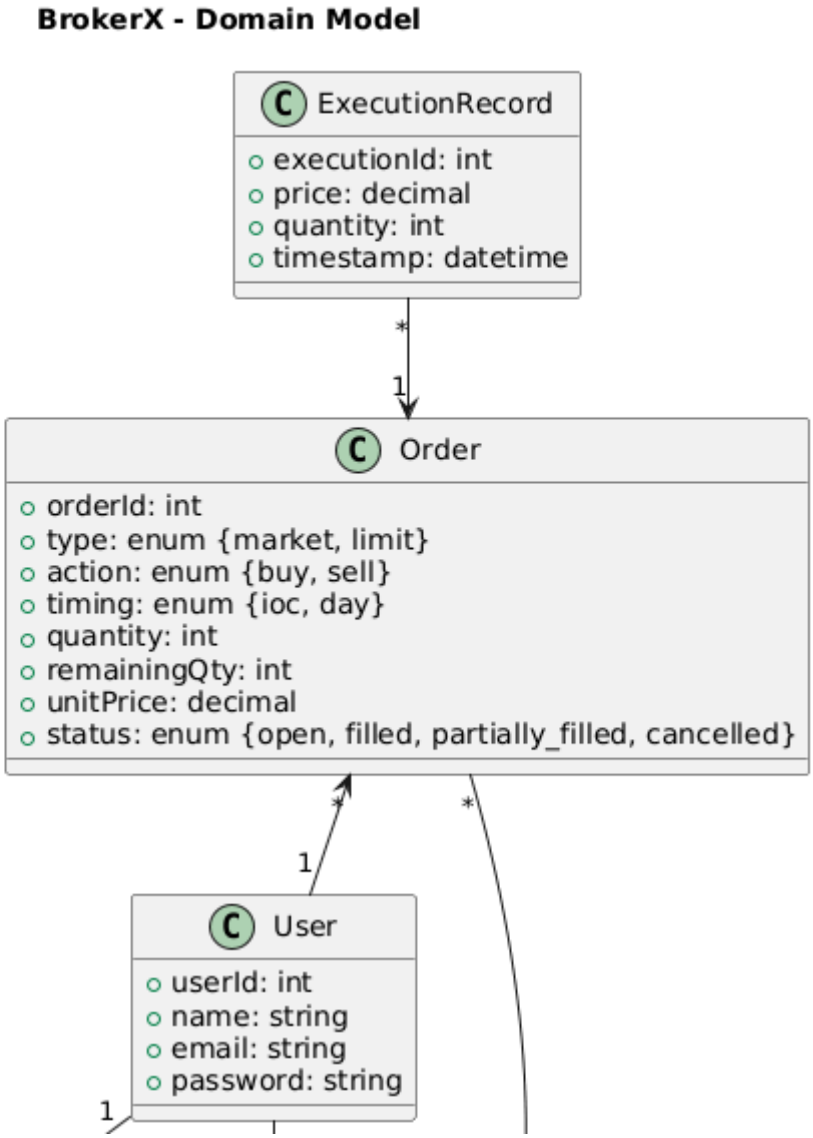
5. Building Block View

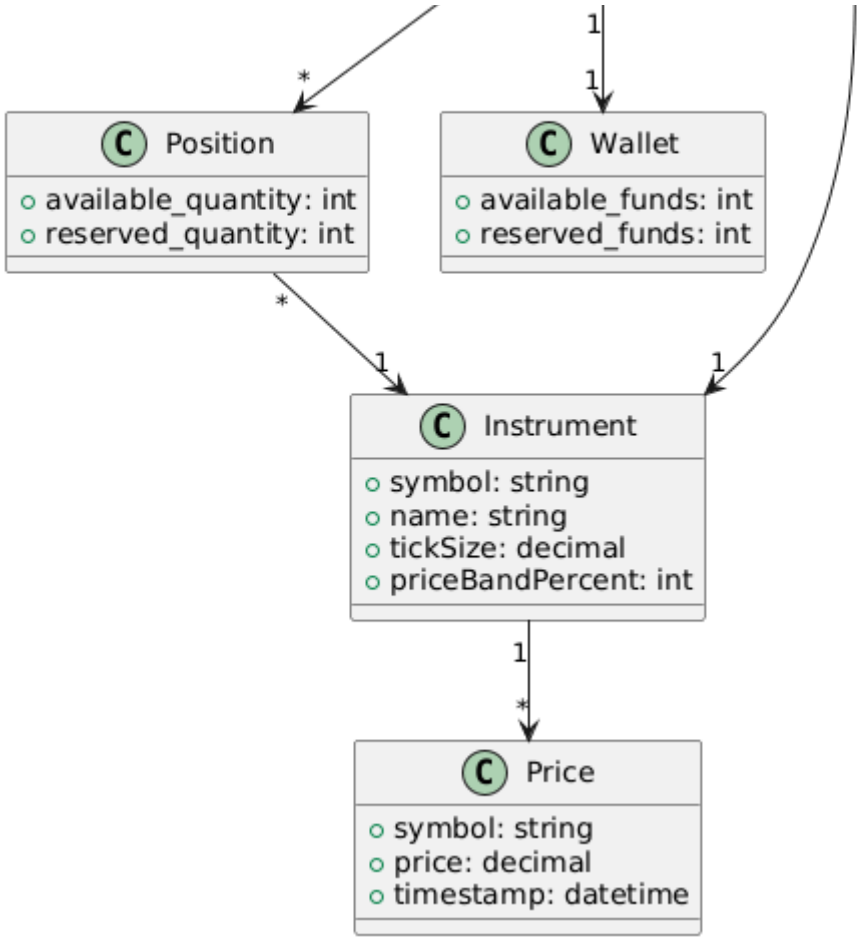



Routes requests to
[HTTP/REST]
«container»
frontend-service
[Go server]
Serves static html javascript files and routes some requests from frontend to backend
Submit/view orders
[HTTP/REST]
Routes requests to
[HTTP/REST]
Routes login and register requests to
[HTTP/REST]
«container»
notification-service
[Go Application]
Handles sending notifications to user
«container»
order-service
[Go RESTful API]
Handles creating and fetching orders
«container»
market-data-service
[Go RESTful API]
Returns market data like stock prices and instruments
«container»
user-service
[Go RESTful API]
Handles authentication and user registration
Consumes Events from
[kafka binary/TCP]
Produces/Consumes Events to/from
[kafka binary/TCP]
Routes requests to
[HTTP/REST]
«container»
Kafka Event Broker
[Kafka]
Handles delivering of events on multiple topics
Produces/Consumes Events to/from
[kafka binary/TCP]
Produces/Consumes Events to/from
[kafka binary/TCP]
Reads/writes
[TCP/IP]
Reads/writes
[TCP/IP]
«container»
matching-service
[Go Application]
Handles order matching and returning orders to the order book
«container»
portfolio-service
[Go API]
Handles balancing wallets and positions quantities
Reads/writes
[RESP/TCP]
Reads/writes
[TCP/IP]
«container»
Redis storage
[Redis]
Stores orders temporarily for order matching
«container»
MySQL Database
[MySQL]
Stores users, wallets, orders, executions, positions persistently


BrokerX - Service level diagram
brokerx
matching
portfolio
market-data
order
user

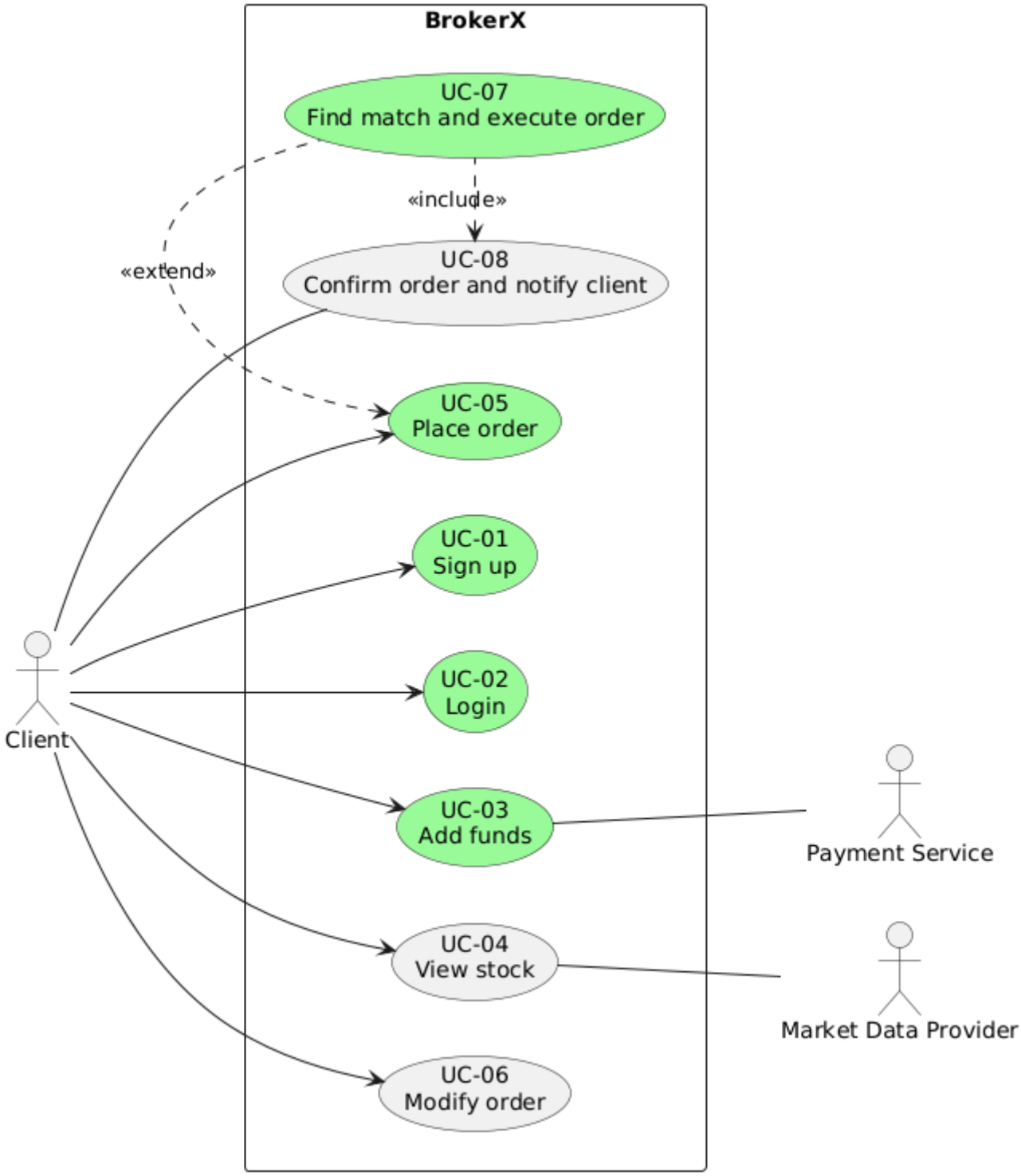

BrokerX - Package level diagram
{service}
adapters
dao
SQL repos and Redis access objects
clients
Kafka event producers
Http clients for calling external service
handlers
Kafka topic consumers and event handlers
Http handlers for request endpoints
core
All internal service (business logic)
implements
implements
implements
references
ports
All external interfaces
uses
references
models
Order
OrderEvent
ClaimedCandidate
OutboxRecord
Instrument
Position
User
Wallet
ExecutionRecord


BrokerX - Domain Model
ExecutionRecord
executionId: int
price: decimal
quantity: int
timestamp: datetime
*
1
Order
orderId: int
type: enum {market, limit}
action: enum {buy, sell}
timing: enum {ioc, day}
quantity: int
remainingQty: int
unitPrice: decimal
status: enum {open, filled, partially_filled, cancelled}
*
1
*
User
userId: int
name: string
email: string
password: string
1



6. Runtime View

General Use Case Diagram

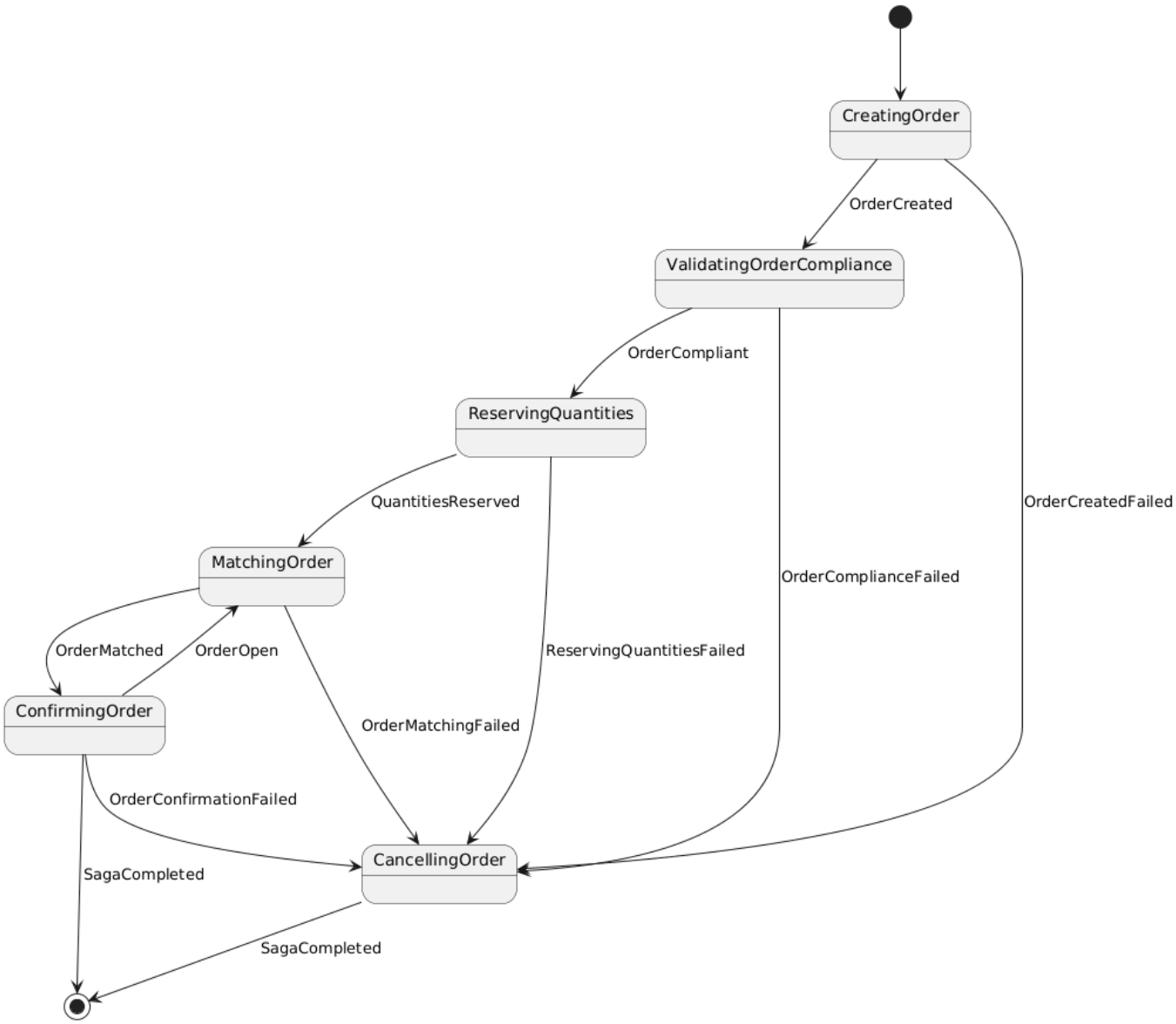


> The use cases colored in green are the ones that are currently implemented in the system

> Note that not all use case scenarios and runtime diagrams have been updated for the phase 3 demo due to lack of time. They will be updated before submission.

Order Saga State Transition Diagram

UC-07 Find match and execute order and UC-08 Confirm order and notify both follow this state transition diagram :



UC-01 Sign up

Goal:

Allow a registered user to create a user account so that they can later authenticate to BrokerX and use it.

Actors:

- Primary: Client (user)
- Supporting: MySQL DB

Preconditions:

N.A.

Main Flow:

1. The Client submits email, password, first name and last name to the system.

2. The system receives the credentials, validates them and creates a new user in the database.

3. The system returns a confirmation of the user registration to the user.

Postconditions:

- User account is created is the system

Exceptions / Alternatives:

E2. Database unavailable → The system returns an error 500 Internal Server Error.

A1.1 User id header not present in request → The system returns an error 401 Unauthorized.

A1.2 Invalid credentials (wrong JSON formatting) → The system returns an error 400 Bad Request.

A2. User account with given email already exists → The system returns an error 409

UC-02 Login

Goal:

Allow a registered user to securely authenticate into the BrokerX system.

Actors:

- Primary: Client (user)
- Supporting: MySQL DB

Preconditions:

- The user is already registered in the system.
- Credentials (email, password hash) are stored in the database.

Main Flow:

1. The Client submits email and password to the system.

2. The system receives the credentials and validates them.

3. The system returns the home page, along with a signed session cookie to the client

Postconditions:

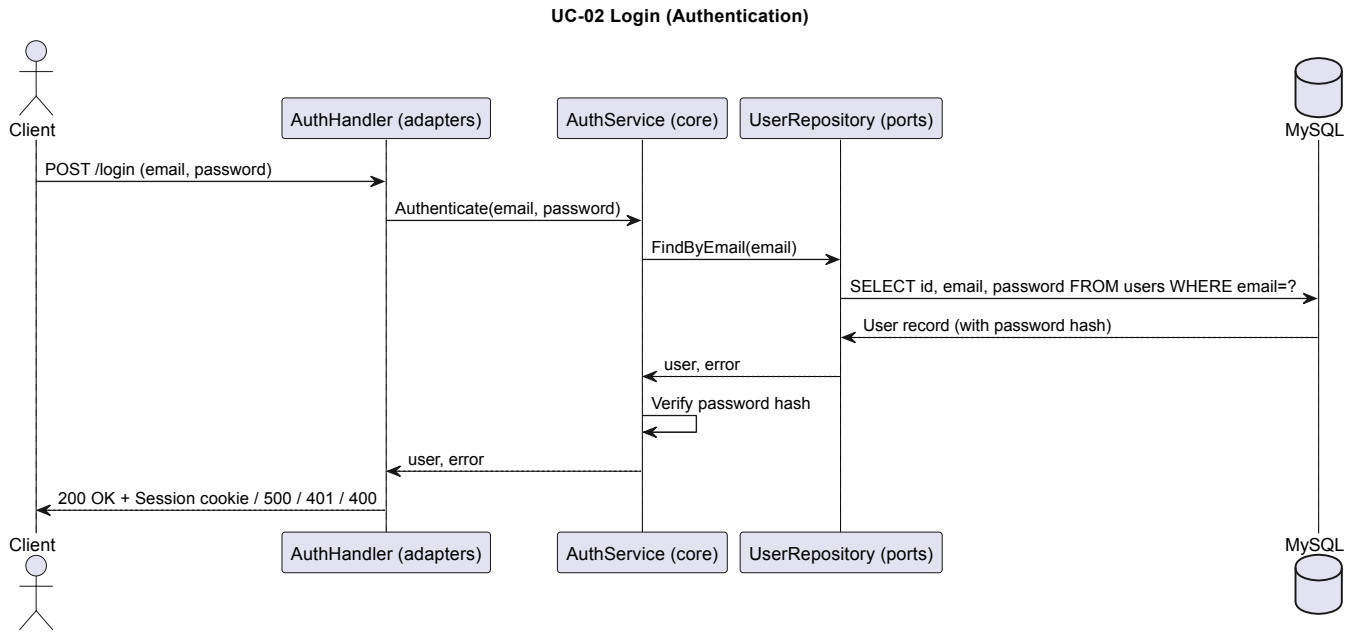
- User is authenticated and can call secured endpoints with the cookie.

Exceptions / Alternatives:

E2. Database unavailable → The system returns an error 500 Internal Server Error.

A3. Invalid credentials → The system returns an error 401 Unauthorized.

A3. Third failed authentication attempt → The system returns an error 401 Unauthorized and locks the account for 30 minutes.



UC-05 Place order

Goal:

Allow a user to place a buy or sell order on a stock.

Actors:

- Primary: Client (user)
- Supporting: Market Data Provider

Preconditions:

- The user is logged in.
- The Market Data Provider is up and running.
- The client is on the Orders page

Main Flow:

1. The Client completes the form to place an order containing the symbol, quantity, action, type, timing and unit price.

2. The system receives the order data and validates that the inputs are not empty.

3. The system starts verification the compliance of the order.

4. The system requests instrument and price data from the Market Data Provider.

5. The Market Data Provider responds with instrument tick size and price band.

6. The system finishes verifying the compliance of the order.

7. The system creates an order.

8. The system submits the order to the internal matching engine.

9. The system send an acknowledgement to the client that the order has been placed.

10. The client receives the acknowledgement.

Postconditions:

- The order is stored in the database with open status.
- The client can see the new placed order with all its information.

Exceptions / Alternatives:

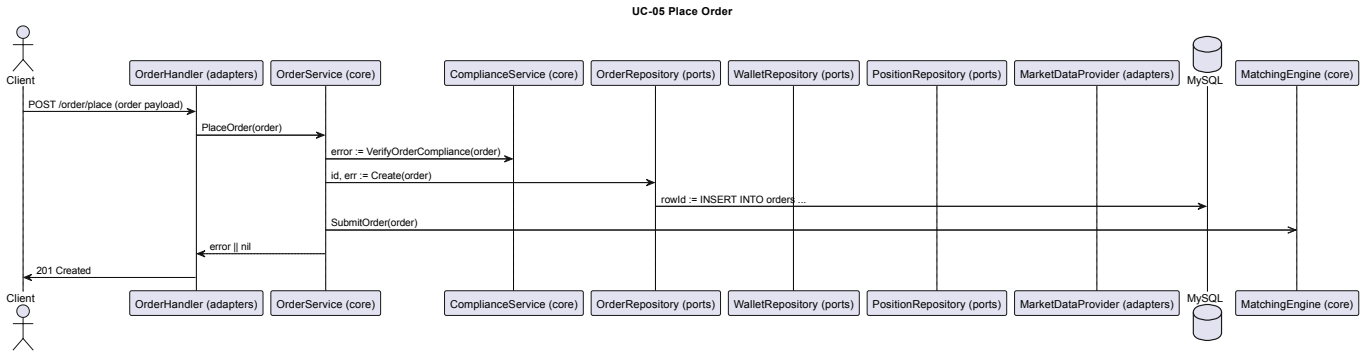
E3. The order has an invalid quantity → The system propagates the error back to the client. The order is not created.

E5. The Market Data Provider does not respond → The system propagates the error back to the client. The order is not created.

E5. The user does not have enough funds → The system propagates the error back to the client. The order is not created.

E5. The user does not have enough stocks → The system propagates the error back to the client. The order is not created.

E5. The order tick size or price band does not match the instrument's required tick size or price band → The system propagates the error back to the client. The order is not created.



UC-07 Find match and execute order

Match incoming buy and sell orders and generates execution records.

Actors:

- Primary: Client (user)
- Supporting: *None*

Preconditions:

- A valid order was placed by the client and submitted to the internal matching engine

Main Flow:

1. The matching engine receives the order.

2. The matching engine fetches all matching orders (symbol, action, type) from the database.

3. The matching engine finds a match or multiple matches for the submitted order.

4. The matching engine generates execution records for each match order.

5. The matching engine updates order(s) quantities and status.

5. The matching engine saves the execution records to the database.

Postconditions:

- The order changes are saved to the database.
- The client can see the submitted order with all its information and the status and quantities updated.
- The client can see each execution record for the order

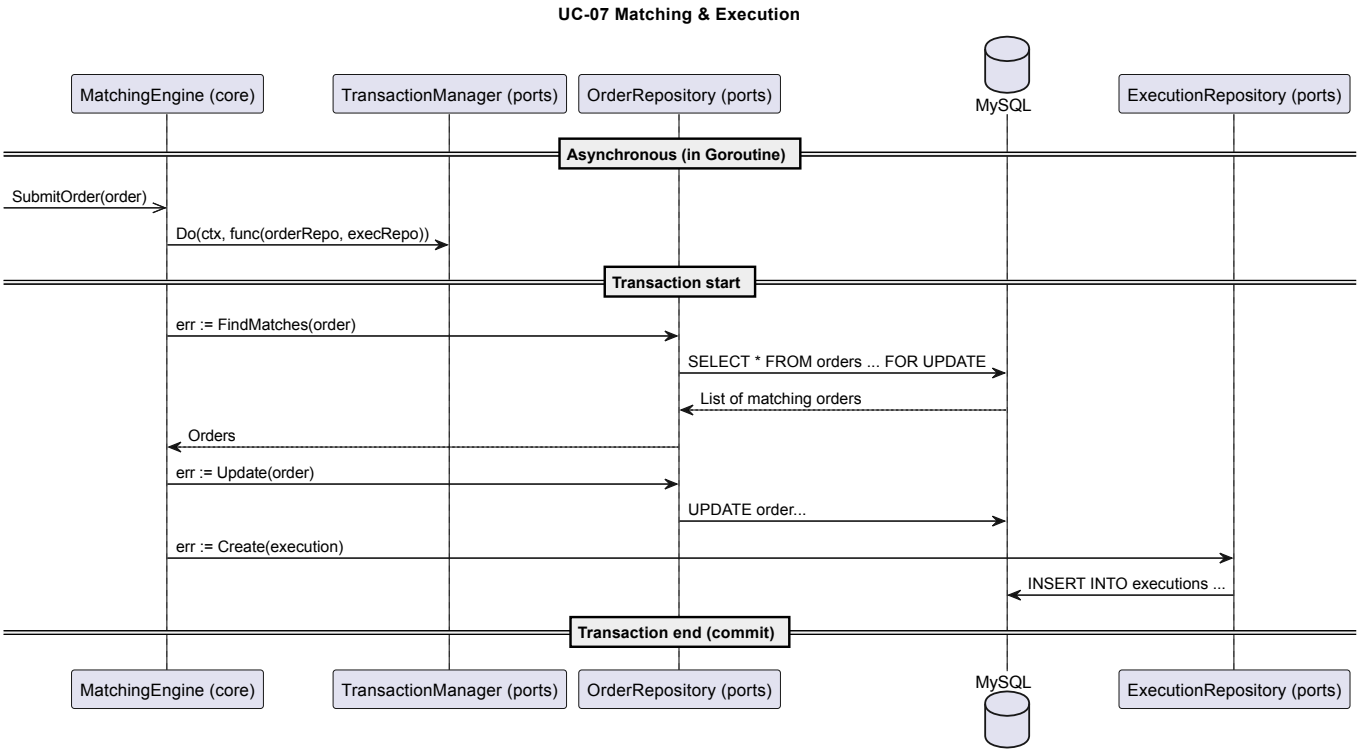
Exceptions / Alternatives:

E2. There is a database error when fetching the orders → The system logs the error. The order stays in the database. The order is not updated.

A2. The systeme finds no match for the order → The order stays in the database. The order is not updated. No execution records are saved.

E5. There is a database error when updating the order(s) → The system logs the error. The order stays in the database. The order is not updated. The execution records are not saved.

E6. There is a database error when saving the execution records → The system logs the error. The order stays in the database. The order is not updated. The execution records are not saved.



UC-08 Confirm order and notify

Update an incoming order and all its claimed candidates's respective user wallets and positions and send an email notification to the users.

Actors:

- Primary: Client (user)
- Supporting: *None*

Preconditions:

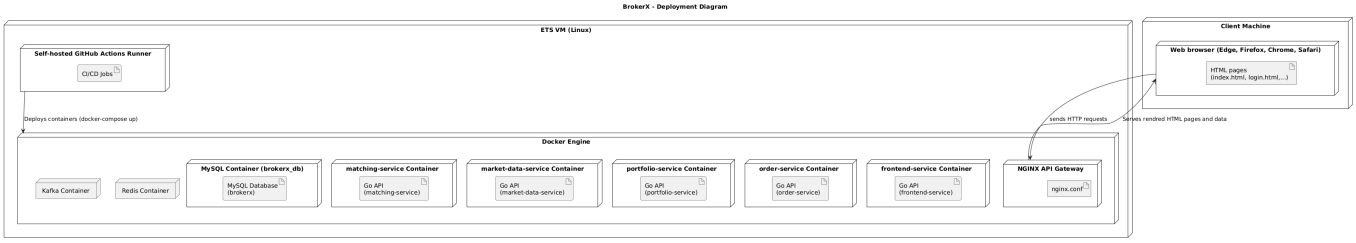
Main Flow:

Postconditions:

Exceptions / Alternatives:

 SVG Image

7. Deployment View



8. Cross-cutting Concepts

- Architecture semi-hexagonale
- Micro-services
- Server-side HTML rendring
- Interfaces
- MySQL relational database
- Repository pattern
- Transaction pattern

9. Design Decisions

ADR-01: Hexagonal architecture

Context

BrokerX must be evolvable across phases: from monolithic prototype to micro-services and eventually event-driven. A tightly coupled MVC design would limit flexibility. We need an architecture that clearly separates domain logic from infrastructure to make refactoring manageable.

Decision

Adopt a hexagonal style architecture (ports and adapters):

- **Core domain layer**: entities, matching engine, business rules
- **Ports**: interfaces for any internal/external service, data access objects
- **Adapters**: implementations of the interfaces (REST API handlers, SQL repositories, mock data provider)

Status

Accepted

Consequences

- Domain logic is independent of delivery and persitence mechanisms
- Easier to swap infrastructure (database, APIs) without touching core logic
- Simplifies testing (domain logic can be tested in isolation)
- Slightly higher complexity (more abstractions, interfaces, files)
- May be over engineering for phase 1 but will most likely pay off in later phases

---

ADR-02: Persistence with MySQL and Repository/Transaction Manager Pattern

Context

BrokerX must maintain strong consistency for orders, executions and balances. Persistent storage was required by the project and we want to avoid spreading raw SQL queries across the codebase.

Decision

Use MySQL relational database as the system of record, accessed through repository interfaces in the ports layers. Repositories abstract database-specific logic into adapters. Transactions are applied for critical operations (order matching).

Status

Accepted

Consequences

- Centralized persistence logic, easier to maintain
- Enforces data integrity using SQL constraints and transactions
- Repository interfaces decouple domain logic from data access details, allowing easier replacement of persistent storage in the future.
- Less flexibility in schema evolution if requirements change rapidly

---

ADR-03: Use of Go as Implementation Language

Context

The backend must support low latency and high throughput for order placement and matching. It must also be portable across environments (developer laptops, CI/CD pipelines, VMs). The following languages were considered :

- Java/C#
- Python/Node.js
- Rust/C

Decision

Implement BrokerX backend API server in Go (Golang)

Status

Accepted

Consequences

- Very fast compilation and deployment cycle.
- Excellent concurrency support for handling many requests/orders.
- Small memory footprint, portable binaries.
- Clean error handling via multiple return values.
- Newer language, meaning lack of libraries in certain domains

ADR-04: NGINX API Gateway

Context

Decision

Implement API Gateway with NGINX.

Status

Accepted

Consequences

ADR-05: Redi-Based Order Book

Context

Decision

Implement Order Book using Redis Sorted Set.

Status

Accepted

Consequences

ADR-06: Grafana, Loki and Promtail for observability

Context

Decision

Use Grafana in conjonction with Promtail and Loki for logs.

Status

Accepted

Consequences

ADR-07: Transactional Outbox Pattern

Context

Decision

Use a transactional Outbox pattern with MySQL for order confirmation steps.

Status

Accepted

Consequences

10. Quality Requirements

Latency

Throughput

Data Integrity

Security

Testability

Interoperability

11. Risks and Technical Debts

- Writing a lot of code in a short amount of time can lead to lack of unit testing and therefore, quality degradation.
- Lack of code review due to the project being individually developped.

- There is a risk of long refactoring time between phases depending on the evolvability of the system.

12. Glossary

Term	Definition
ADR	Architecture Decision Record: a short document that captures an important architectural choice, its context, and consequences.
Broker	An intermediary that executes buy/sell orders on behalf of investors. BrokerX acts as an online broker for retail clients.
CI	Continuous Integration: automation of integrating new code into the main branch with quality checks and automated tests.
CD	Continuous Deployment: automation of delivering tested code to production environments.
Execution	The successful match of a buy and sell order, resulting in a trade and an execution report.
Execution Report	A message generated by the matching engine that details the outcome of an order execution (price, quantity, time).
Hexagonal Architecture	Also called Ports and Adapters: an architectural style that separates the domain logic (core) from external systems (DB, UI, APIs).
Latency	The time it takes for a user action (e.g., placing an order) to receive an acknowledgement from the system.
Limit Order	An order to buy or sell a stock at a specified price or better.
Market Data Provider	External or simulated service that supplies price and instrument data to BrokerX.
Market Order	An order to buy or sell immediately at the current market price.
Matching Engine	The core system component that pairs buy and sell orders and generates execution records.
Monolith	An application built as a single deployable unit. BrokerX phase 1 is implemented as a Go monolith.
MySQL	Relational database used by BrokerX to persist orders, executions, users, and balances.
Order	A request placed by a user to buy or sell a stock (can be market or limit, buy or sell, ioc or day).
IOC	Immediate Or Cancel. Refers to the timing of an order that must be immediately fully filled, if not, it must be cancelled
DAY	Refers to the timing of an order that is open until the end of the trading day.

Term	Definition
Order Book	The collection of all open buy and sell orders for a given instrument, managed by the matching engine.
Port	Interface in hexagonal architecture that represents an entry/exit point for the core domain logic (e.g., repository interface, service interface).
Adapter	Implementation of a port, connecting the domain logic to infrastructure (e.g., SQL repository, HTTP services).
Pre-trade Checks	Compliance rules applied before accepting an order (e.g., tick size validation, price bands, sufficient balance/holdings).
Session Cookie	Small piece of data stored on the client after login, used to authenticate subsequent requests.
Tick Size	The minimum price movement allowed for a given instrument.
Throughput	Number of orders the system can successfully handle per second.
Transaction	A group of one or more database operations executed as a single unit of work, ensuring atomicity and consistency.
Wallet	Virtual balance associated with a user account, used to fund purchases and receive proceeds from sales.

Performance tests results

During the second phase of the Brokerx project, a myriad of load tests were done to assess the throughput, latency and availability of the app depending on the architecture and configuration. A Locust container was used to send send requests to the BrokerX endpoint responsible for processing orders.

Here are the test results in chronological order with a description of the improvements made at each iteration.

Phase 1: Monolithic BrokerX

No caching, no load balancing

100 users with peak ~65 orders placed/second

Stays usable to whole time, but a slight increase in latency is observed over time.

200 users with peak ~105 orders placed/second

Quickly becomes unusable after reaching 100 RPS due to latency surpassing 500 ms and degrading over time.

> The performance tests above revealed the following issues which will be fixed in the next iterations :
>
> - One goroutine created per request to submit to matching engine (adressed in Introduce caching #5)
> - Requests to fetch orders every time an order is placed (adressed in Introduce caching #5)
> - Heavy load on CPU caused by html template rendering for every single request (adressed in Transform into RESTful API #1)
> - Lack of Db connection pool management (connections not closing, not being reused) (adressed in Introduce caching #5)
> - Heavy logging (adressed in Structure logs #7)

100 users with peak ~65 orders placed/second

The main culprit was found to be the html rendering for every single request. After removing html rendering, the cpu load is constant with constant requests. Image

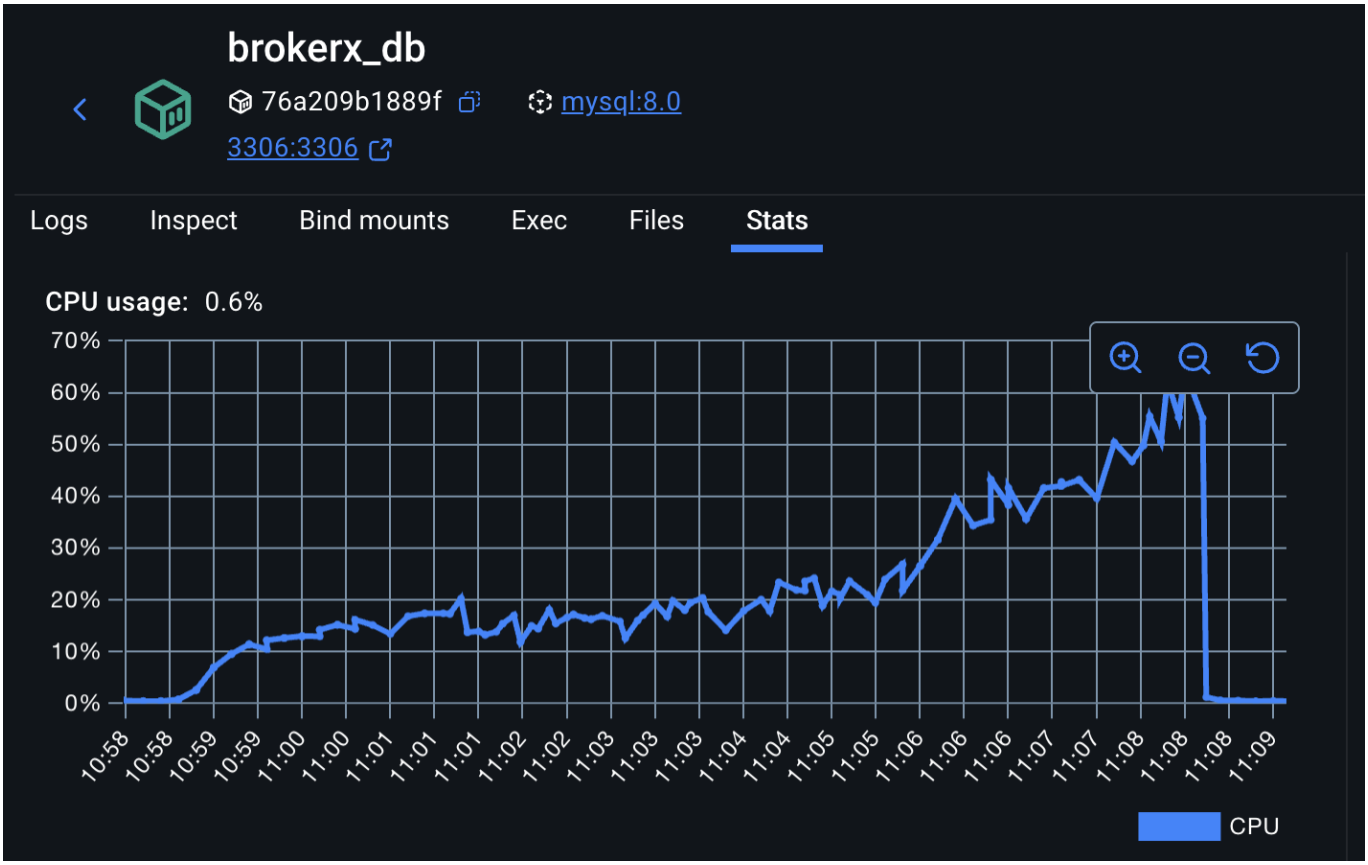
However, the cpu load on the mysql database container is steadily increasing with constant requests. This suggests that order fetching becomes more and more demanding, the more our tables grow. Investigation is underway

> With the current schema creation script, there is no index on the fields used by the matching engine to fetch, making fetching time of orders grow linearly with growing orders table.

> Here are the solutions that encourage a logarithmic increase in database cpu usage (load stabilizes with growing tables):

> - Added two indexes on the orders table CREATE INDEX idx_orders_match_asc ON orders(symbol, action, status, unit_price ASC, created_at ASC); CREATE INDEX idx_orders_match_desc ON orders(symbol, action, status, unit_price DESC, created_at ASC); (To be implemented)
> - Add Redis queue that created orders are sent to and make matching engine read from that queue instead of always reading from database, therefore reducing load on mysql database container

With the two indexes added, we can observe a logarithmic increase of cpu load on the db between 10:59 and 11:06 when the Request load was constant. At 11:06, the RPS was doubled, which explains the sudden rise :



Implemented batch writes for execution records. Performance is stable over time, but db usage still keeps rising the more orders are stored in the orders table.


LOCUST
Host
http://nginx:80
Status
STOPPED
Workers
10
RPS
67.4
Failures
0%
NEW
RESET
STATISTICS
CHARTS
FAILURES
EXCEPTIONS
CURRENT RATIO
DOWNLOAD DATA
LOGS
WORKERS
LOCUST CLOUD
Total Requests per Second
RPS
Failures/s
Response Times (ms)
50th percentile
95th percentile
Number of Users
Number of Users
8:30:00 PM
8:32:00 PM
8:34:00 PM
8:36:00 PM
8:38:00 PM
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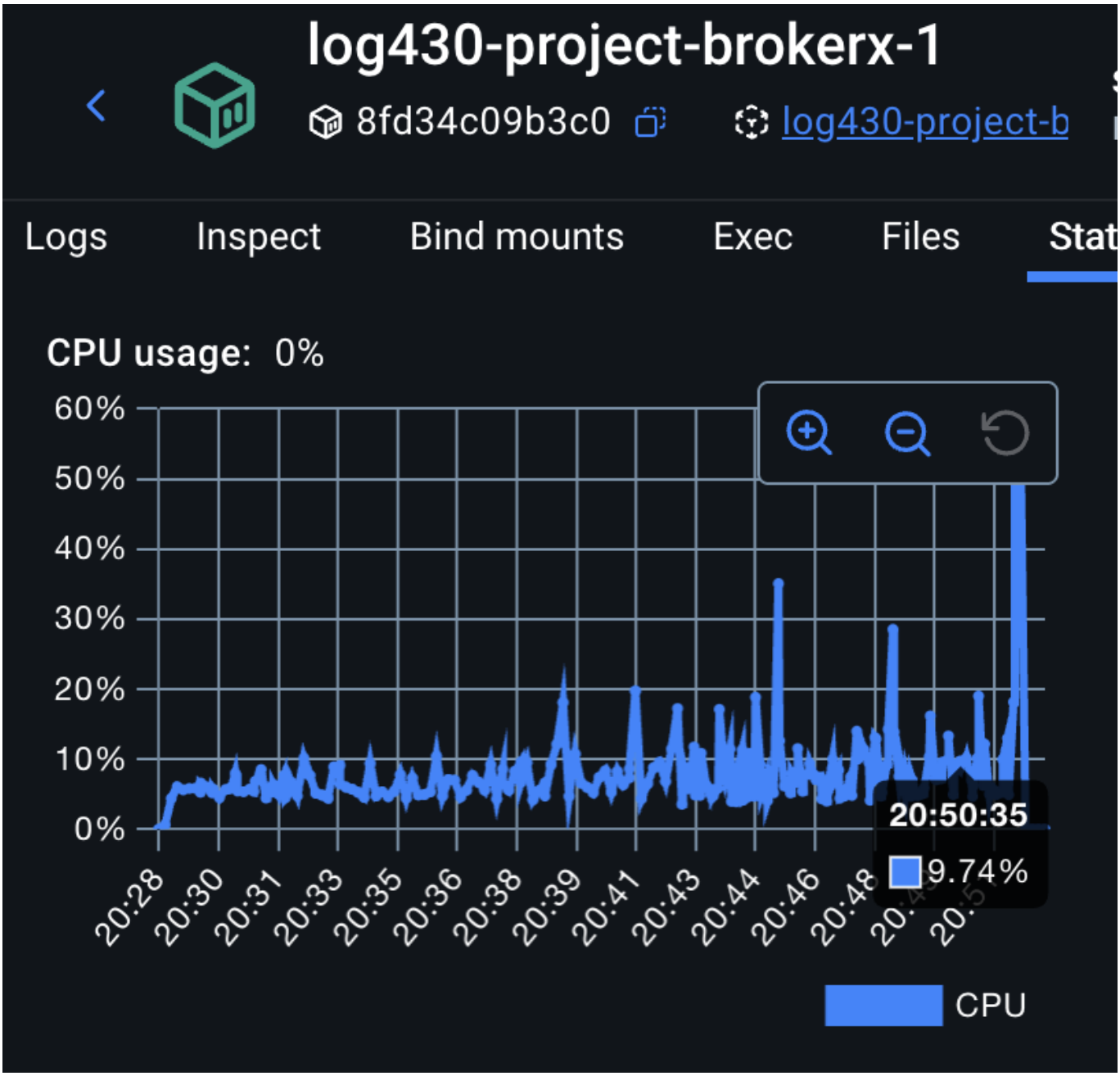
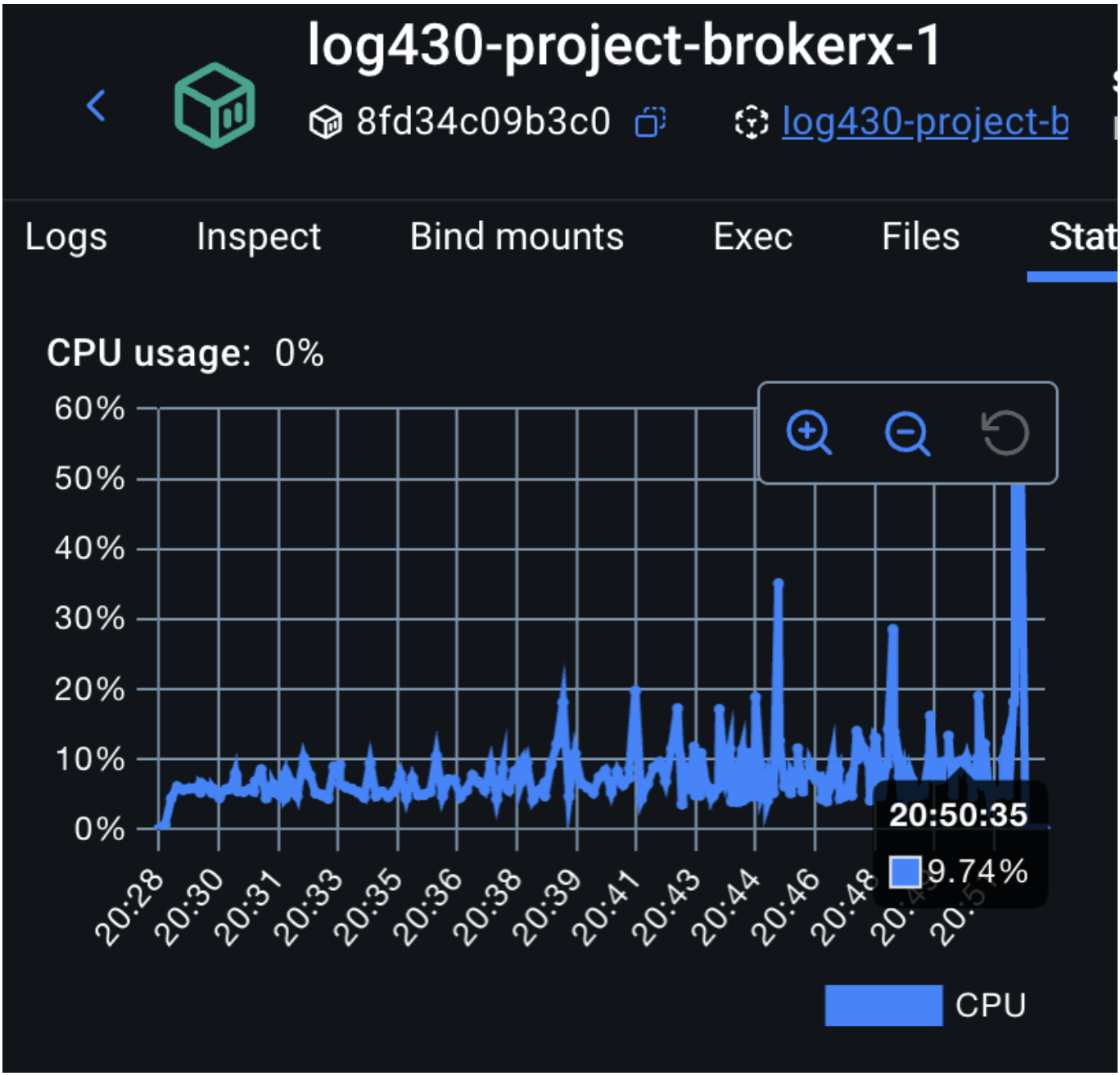
8:48:00 PM
8:50:00 PM
8:52:00 PM


log430-project-brokerx-1
8fd34c09b3c0
log430-project-b
Logs
Inspect
Bind mounts
Exec
Files
Stat
CPU usage: 0%
60%
50%
40%
30%
20%
10%
0%
20:28
20:30
20:31
20:33
20:35
20:36
20:38
20:39
20:41
20:43
20:44
20:46
20:48
20:50:35
9.74%
CPU



Added Redis Order book (caching), no load balancing

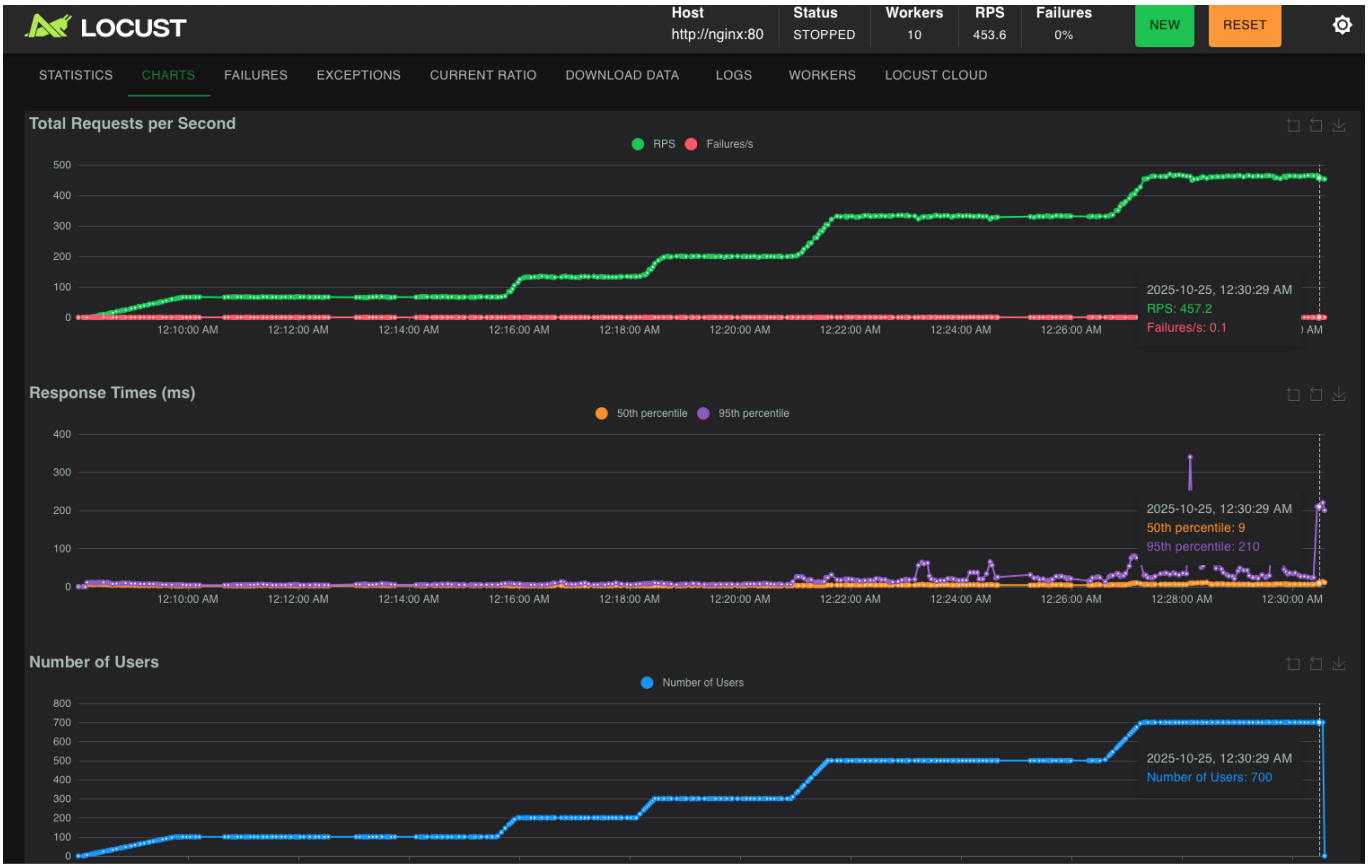
The problem observed in the previous tests was that the mysql database or any relationnal database will be the bottleneck of this application if we continue fetching and creating rows continuously in a growing orders table. It is inevitable. Therefore, the next logical step is using Redis to keep a live in memory order book :

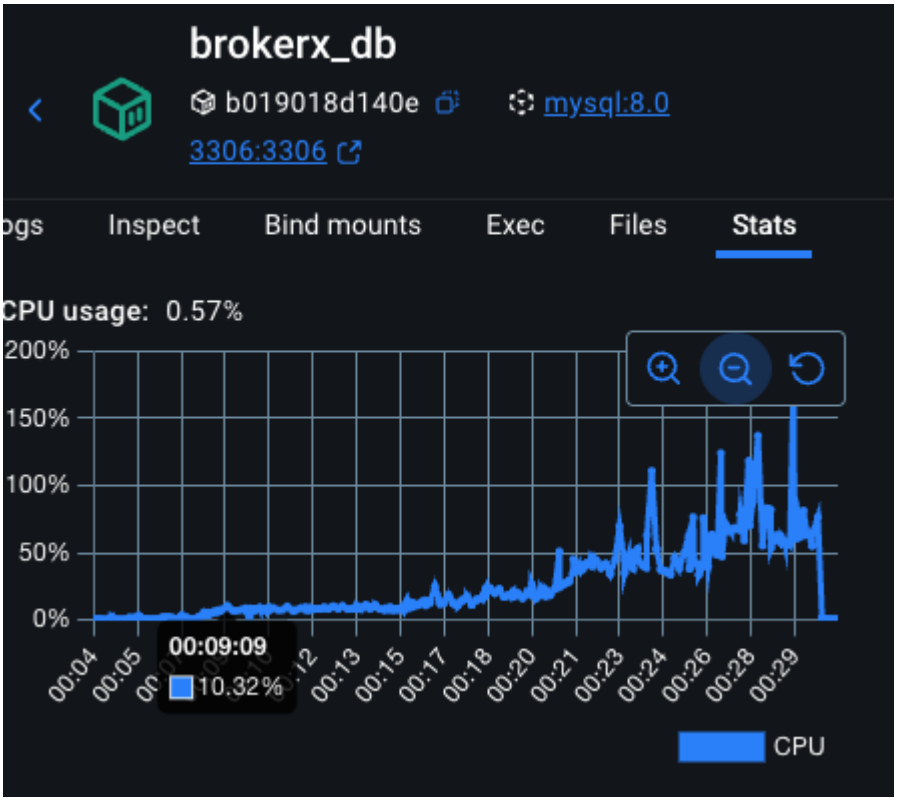
- Redis Sorted Set = “live order book” (fast, concurrent, transient state)
- MySQL = “persistent ledger” (durable, slower, authoritative snapshot)

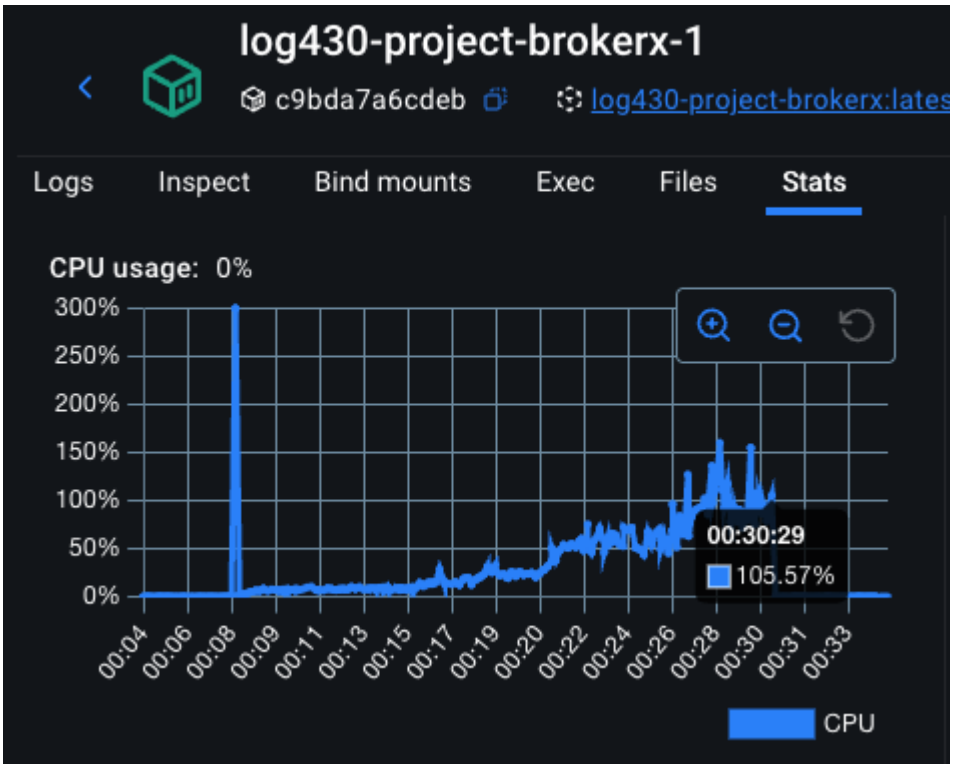
Adding redis to take care of the live order book really helped to stabilize the cpu load on the mysql container since most operations are done using the redis ordered set which is a lot more performant than constant sql queries to fetch and set data.

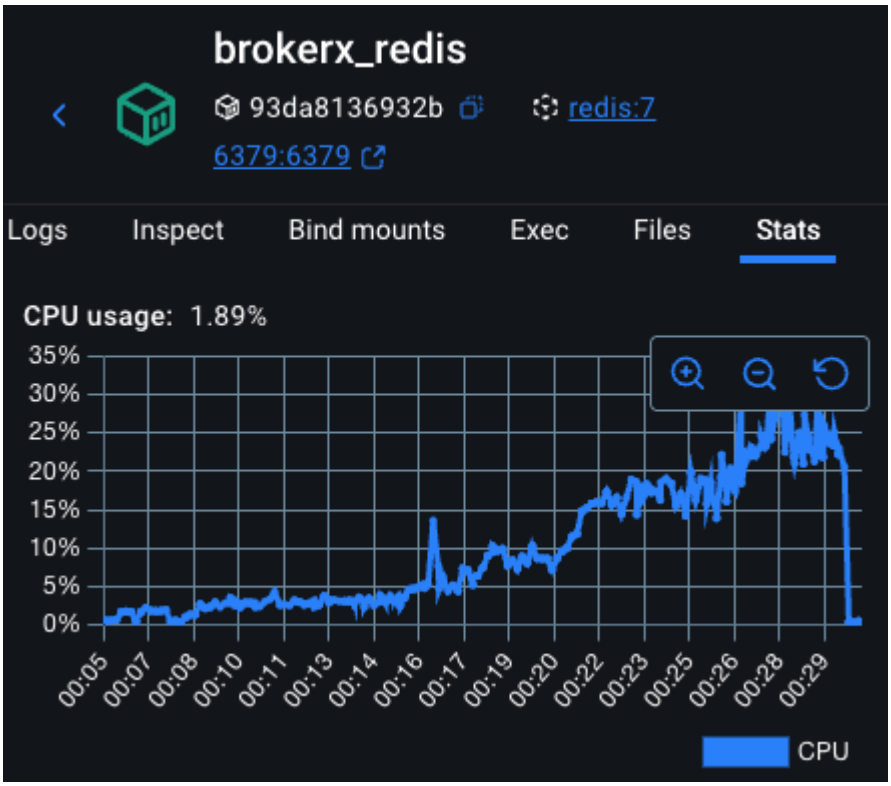
Load test up to 400 RPS

Here are the results of the first performance test after these redis changes :


LOCUST
Host
http://nginx:80
Status
STOPPED
Workers
10
RPS
453.6
Failures
0%
NEW
RESET
STATISTICS
CHARTS
FAILURES
EXCEPTIONS
CURRENT RATIO
DOWNLOAD DATA
LOGS
WORKERS
LOCUST CLOUD
Total Requests per Second
RPS
Failures/s
2025-10-25, 12:30:29 AM
RPS: 457.2
Failures/s: 0.1
Response Times (ms)
50th percentile
95th percentile
2025-10-25, 12:30:29 AM
50th percentile: 9
95th percentile: 210
Number of Users
Number of Users
2025-10-25, 12:30:29 AM
Number of Users: 700


brokerx_db
b019018d140e
mysql:8.0
3306:3306
Inspect
Bind mounts
Exec
Files
Stats
CPU usage: 0.57%
200%
150%
100%
50%
0%
00:09:09
10.32%
CPU





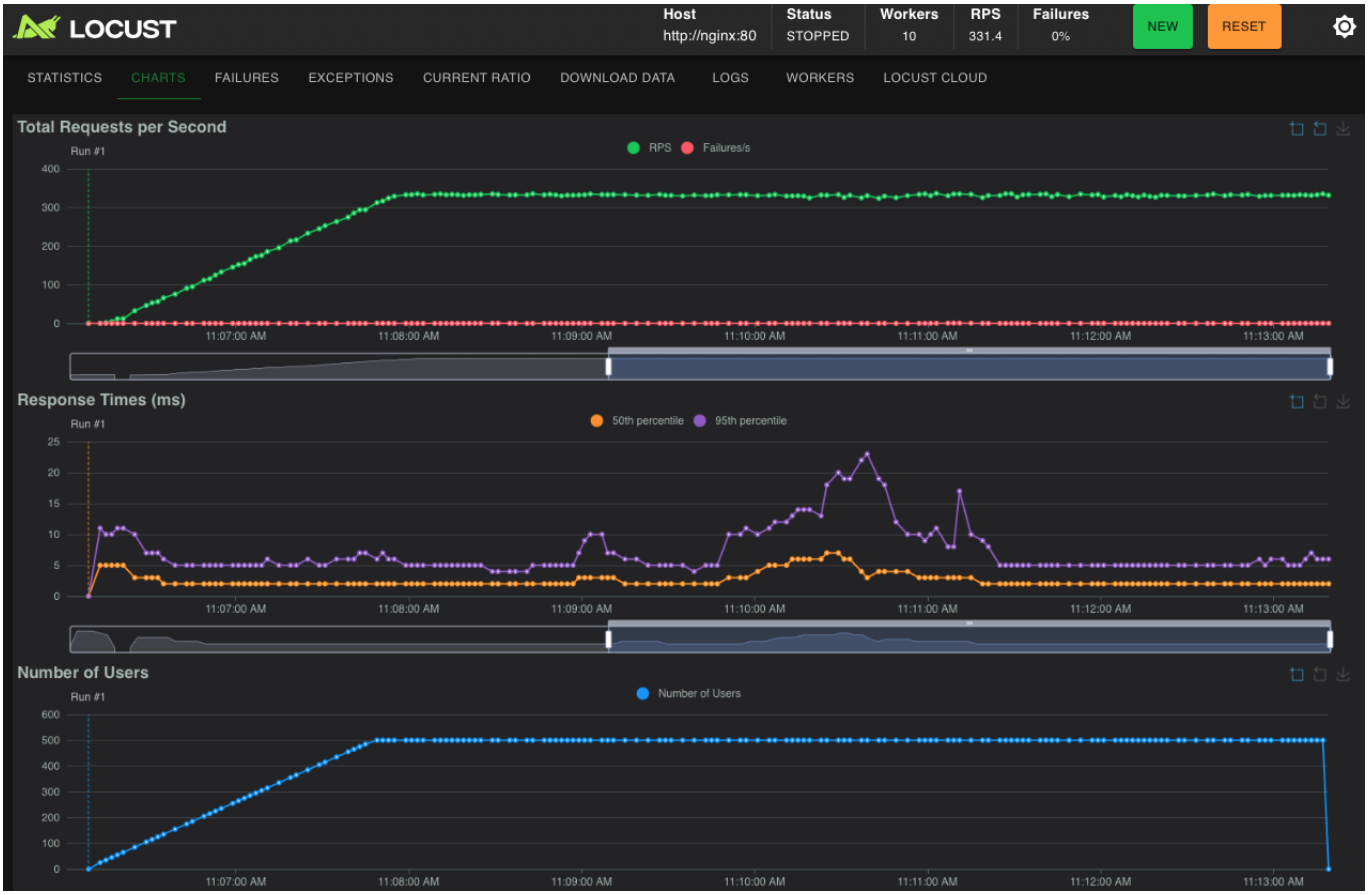
With the load increasing and reaching almost 500 RPS on only 1 instance! It is normal to see the cpu load increasing since the load was also increasing every few minutes. But what is important is that the latency stayed way under the required 500 ms and the cpu load somewhat stabilized after reaching each plateau of request load.

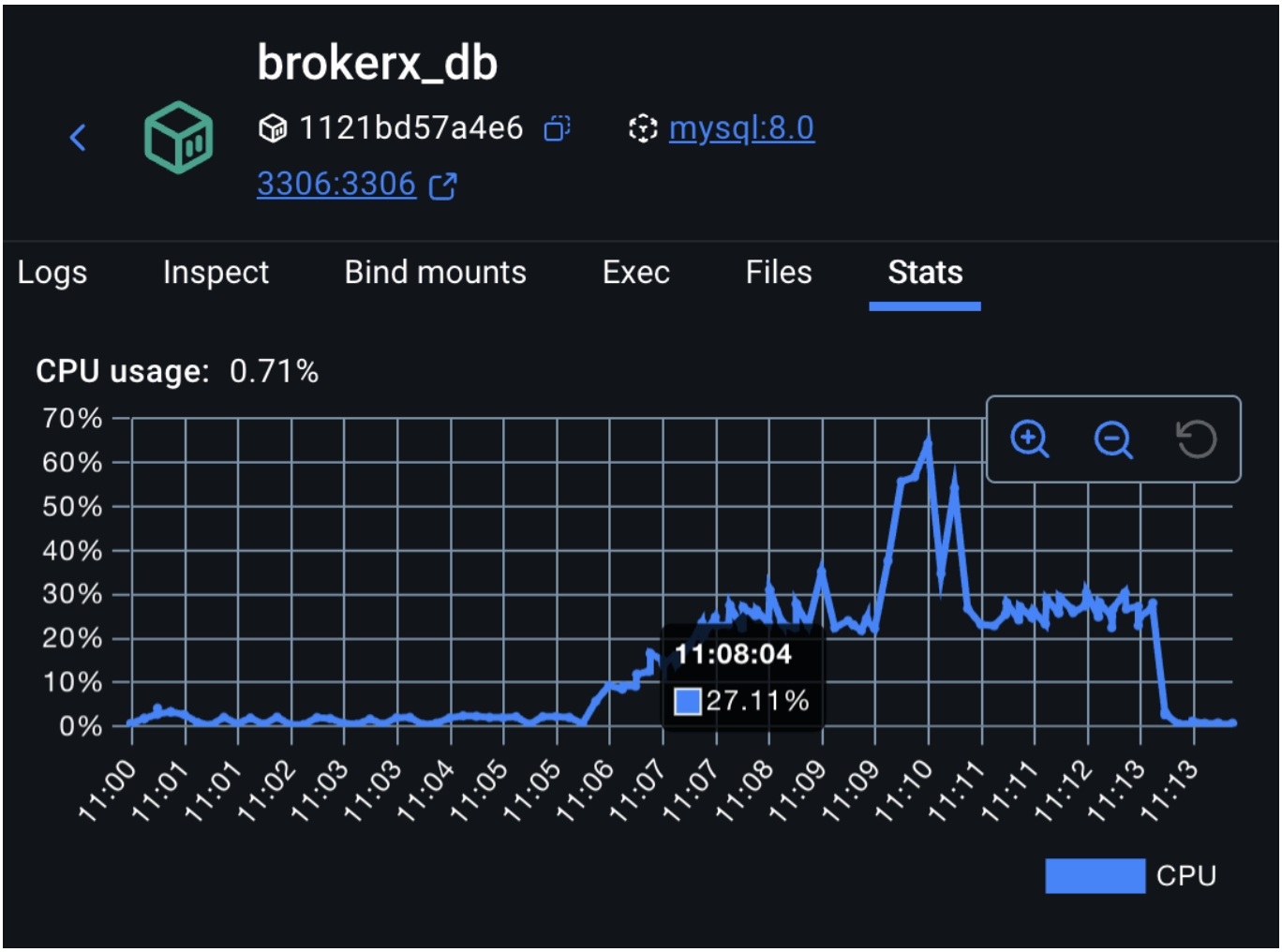
Next steps are :

- Implement regular order syncing from redis to mysql so that incomplete orders are visible to the user
- OPtimize sql performance by changing order and execution persistance by sending to queue and flushing every x seconds

- Remove unecessary composite sql indexes
- Optimize locust performance so that it takes less cpu % to eventually reach 1000 RPS
- Try load balancing go api to reduce stress on each instance
- Try distributed sql database? Or other database (see class slides)

Performance test result after implementing regular dirty order syncing (7 minutes, 500 users, 5 users/second)




brokerx_db
1121bd57a4e6
mysql:8.0
3306:3306
Logs
Inspect
Bind mounts
Exec
Files
Stats
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70%
60%
50%
40%
30%
20%
10%
0%
11:08:04
27.11%
11:00
11:01
11:01
11:02
11:03
11:03
11:04
11:05
11:05
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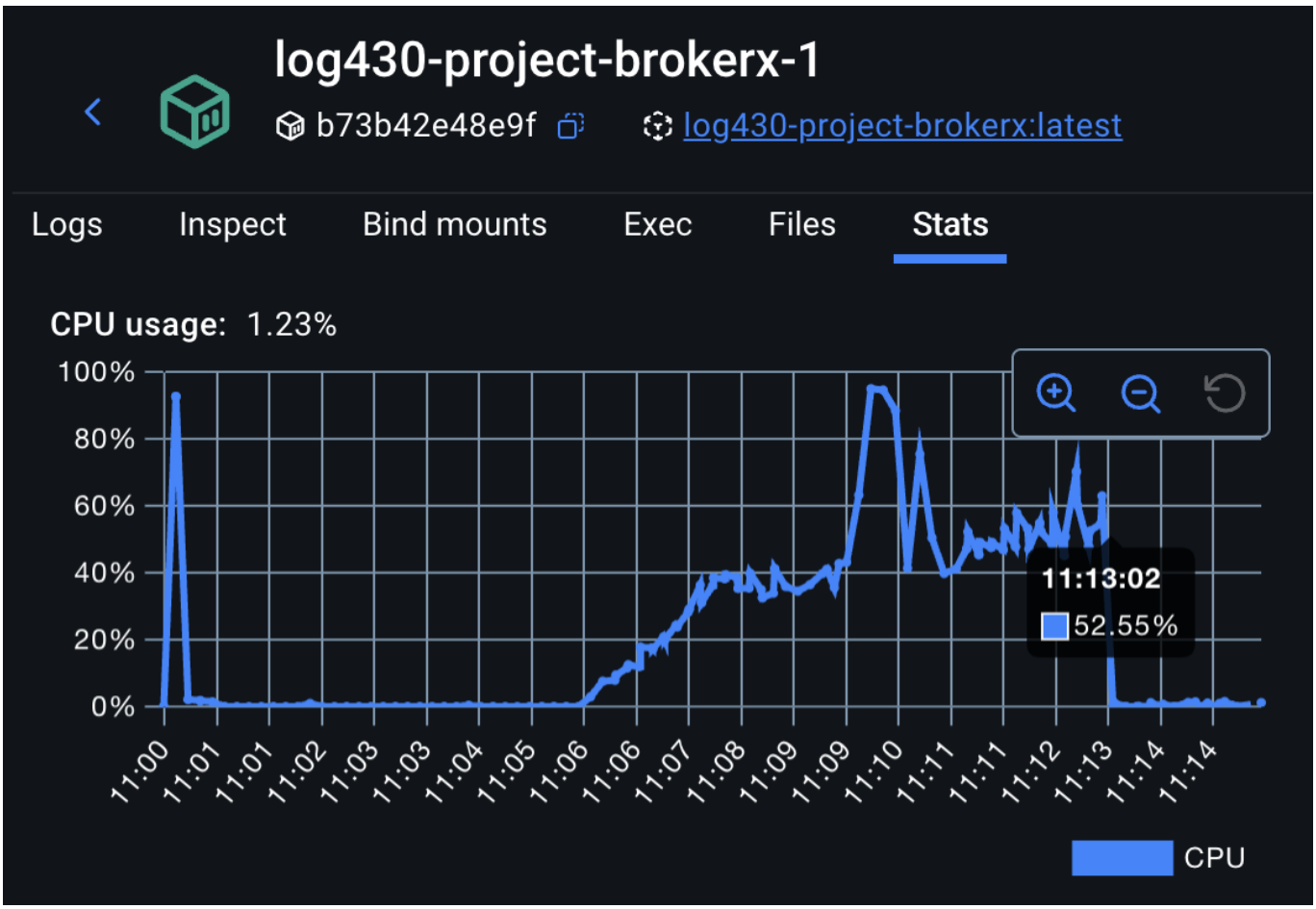
11:10
11:11
11:11
11:12
11:13
11:13
CPU


log430-project-brokerx-1
b73b42e48e9f
log430-project-brokerx:latest
Logs
Inspect
Bind mounts
Exec
Files
Stats
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100%
80%
60%
40%
20%
0%
11:13:02
52.55%
11:00
11:01
11:01
11:02
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11:03
11:04
11:05
11:06
11:06
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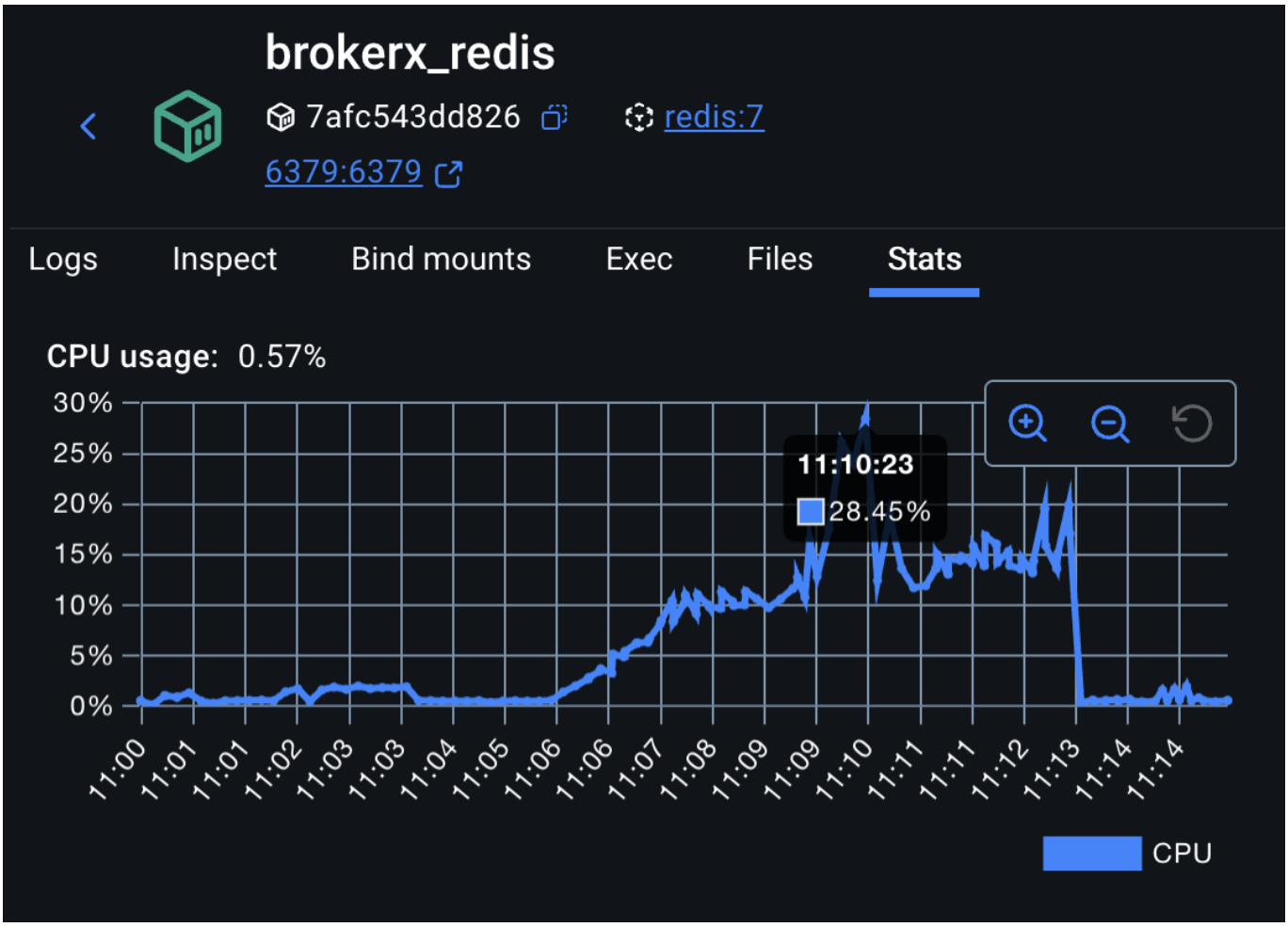
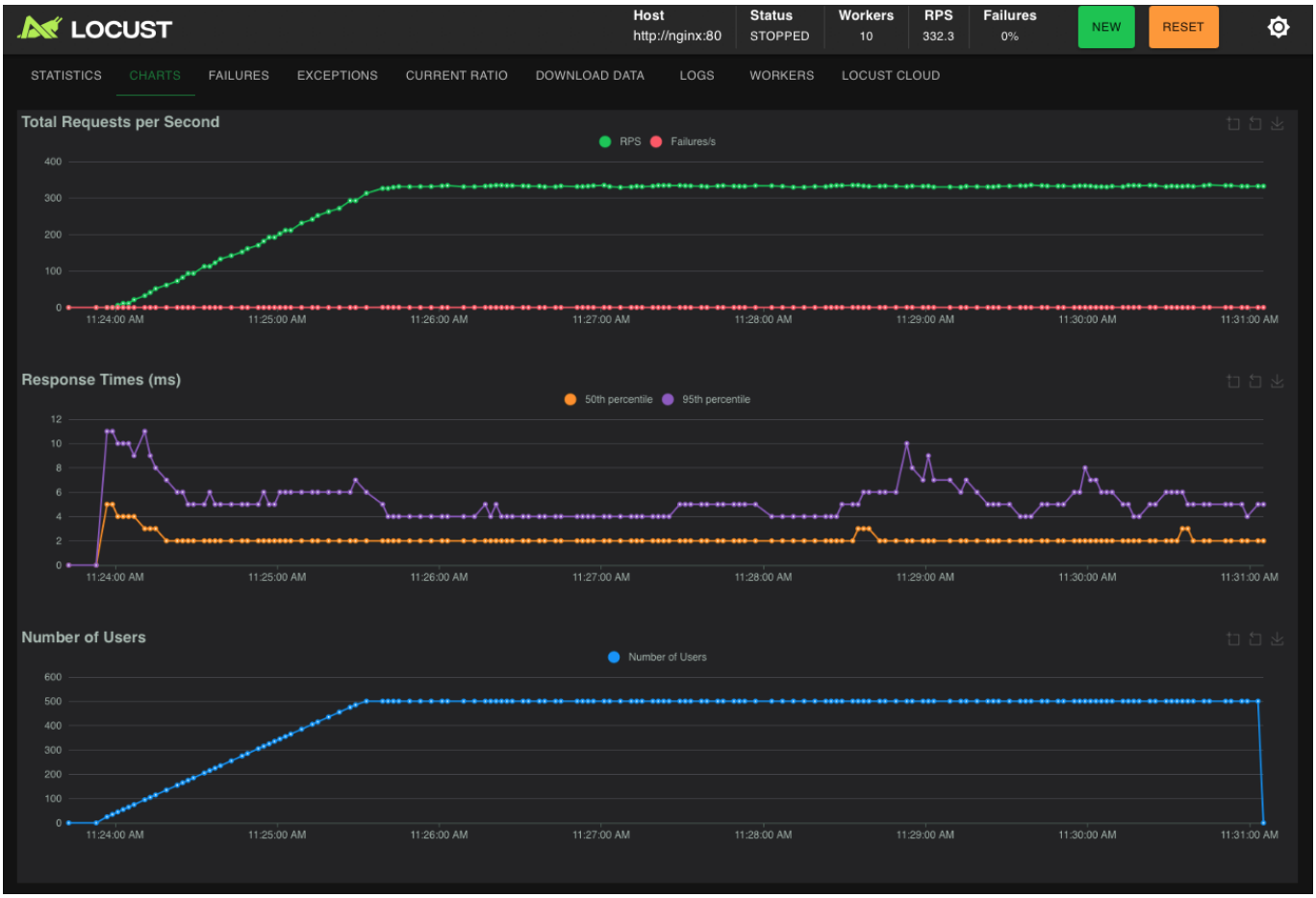
11:12
11:13
11:14
11:14
CPU



Performance test result after removing unused composite indexes (7 minutes, 500 users, 5 users/second)


LOCUST
Host
http://nginx:80
Status
STOPPED
Workers
10
RPS
332.3
Failures
0%
NEW
RESET
STATISTICS
CHARTS
FAILURES
EXCEPTIONS
CURRENT RATIO
DOWNLOAD DATA
LOGS
WORKERS
LOCUST CLOUD
Total Requests per Second
RPS
Failures/s
Response Times (ms)
50th percentile
95th percentile
Number of Users
Number of Users
11:24:00 AM
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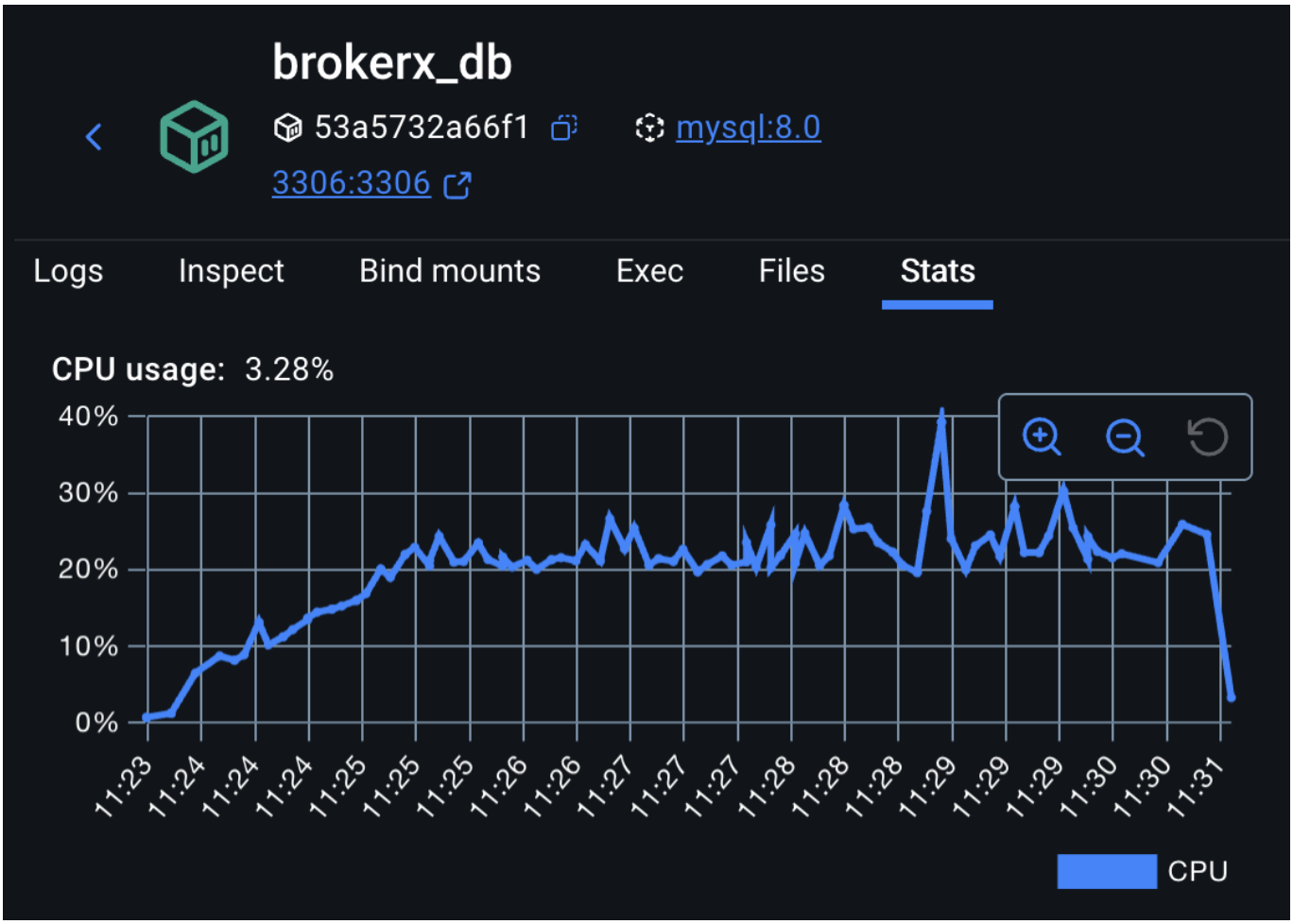
11:27:00 AM
11:28:00 AM
11:29:00 AM
11:30:00 AM
11:31:00 AM


brokerx_db
53a5732a66f1
mysql:8.0
3306:3306
Logs
Inspect
Bind mounts
Exec
Files
Stats
CPU usage: 3.28%
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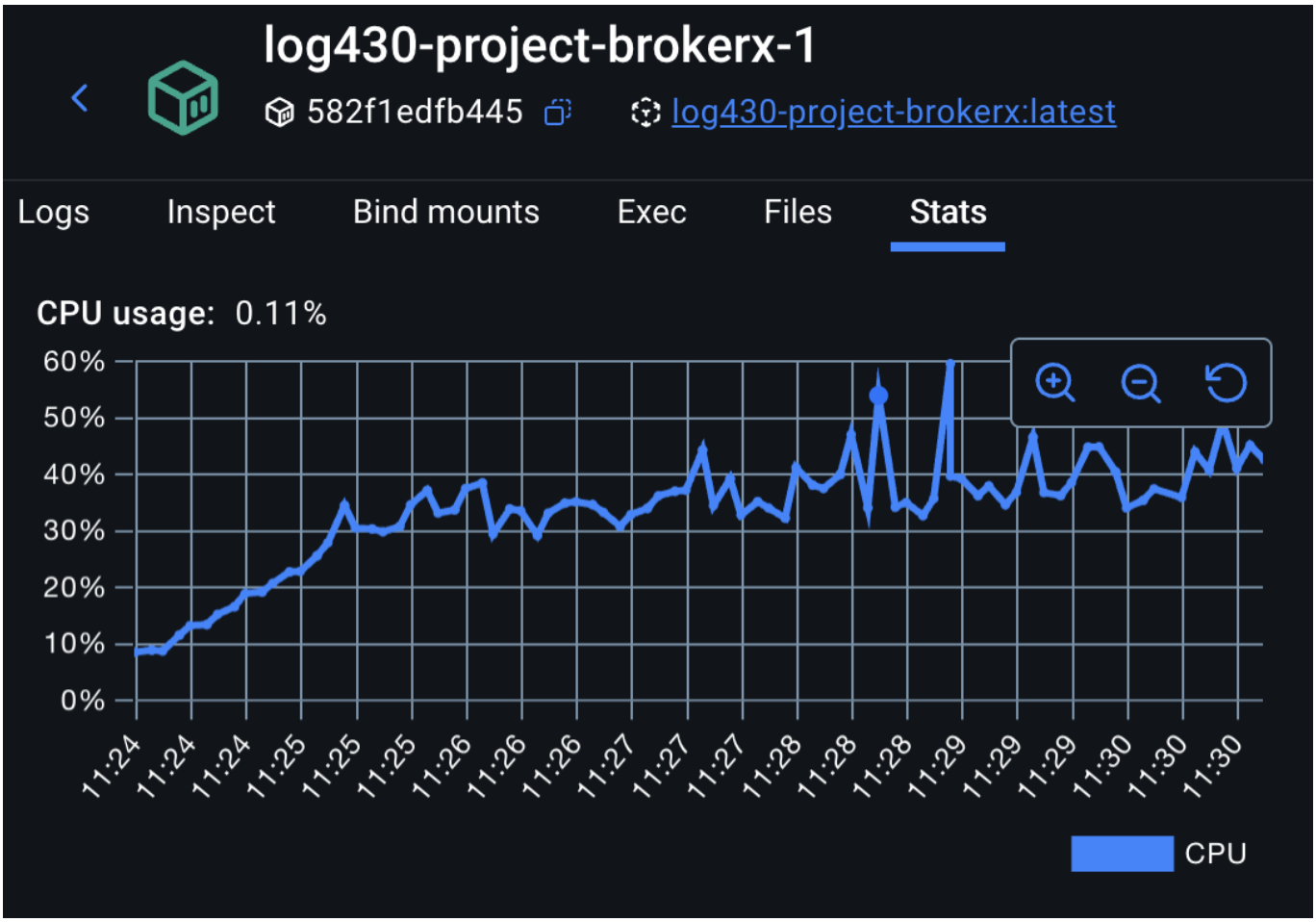
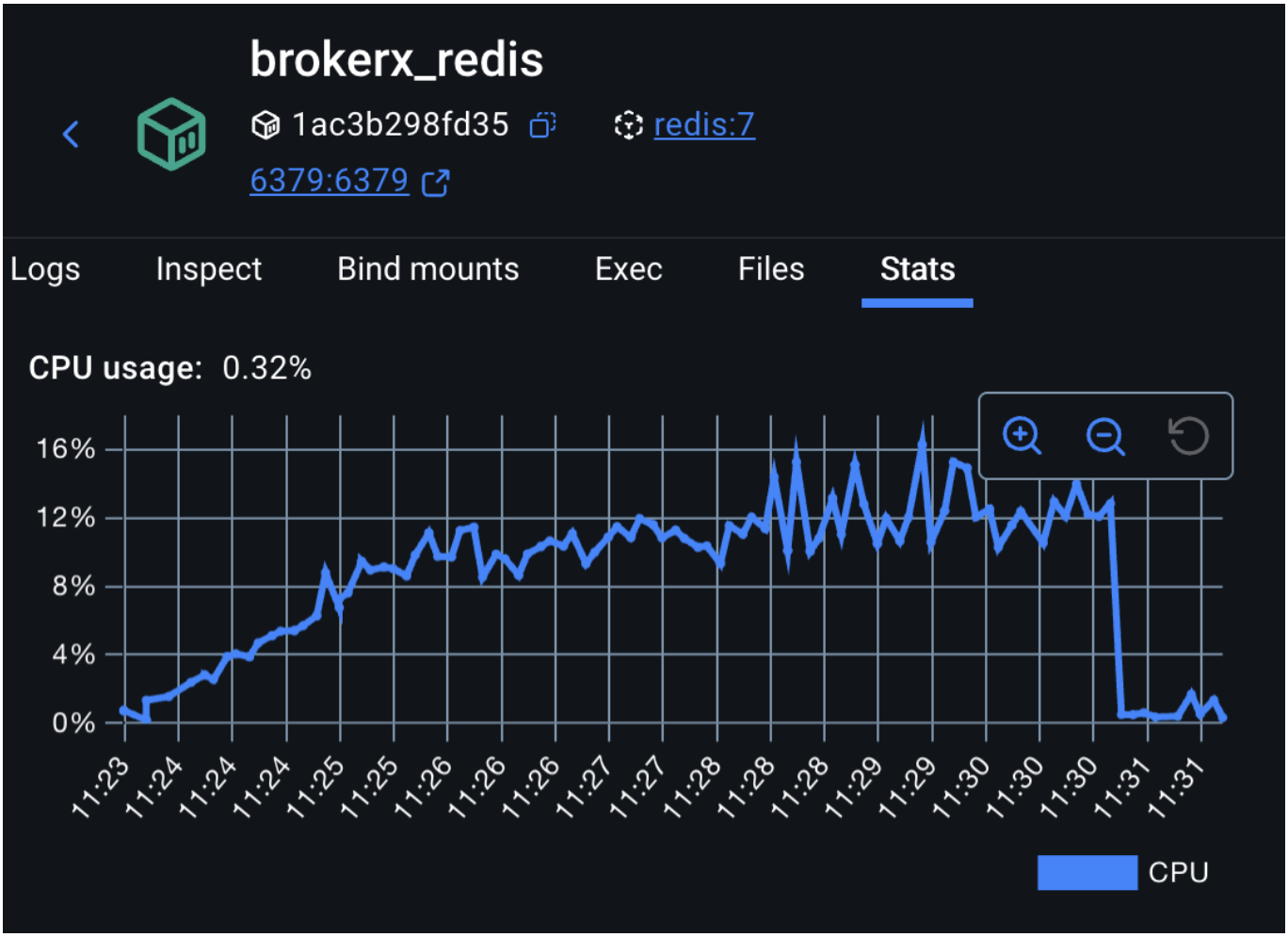
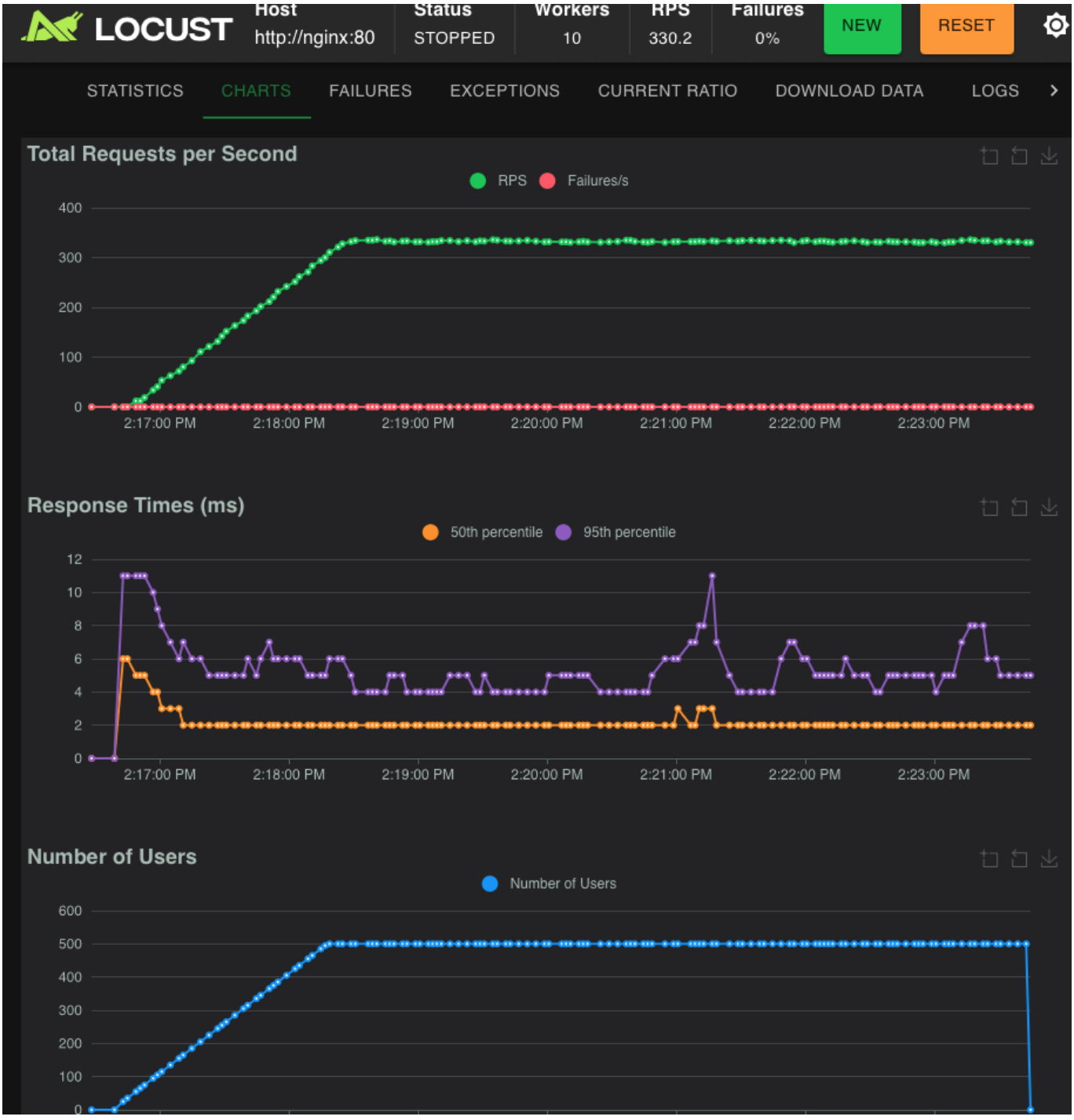
11:30
11:31
CPU

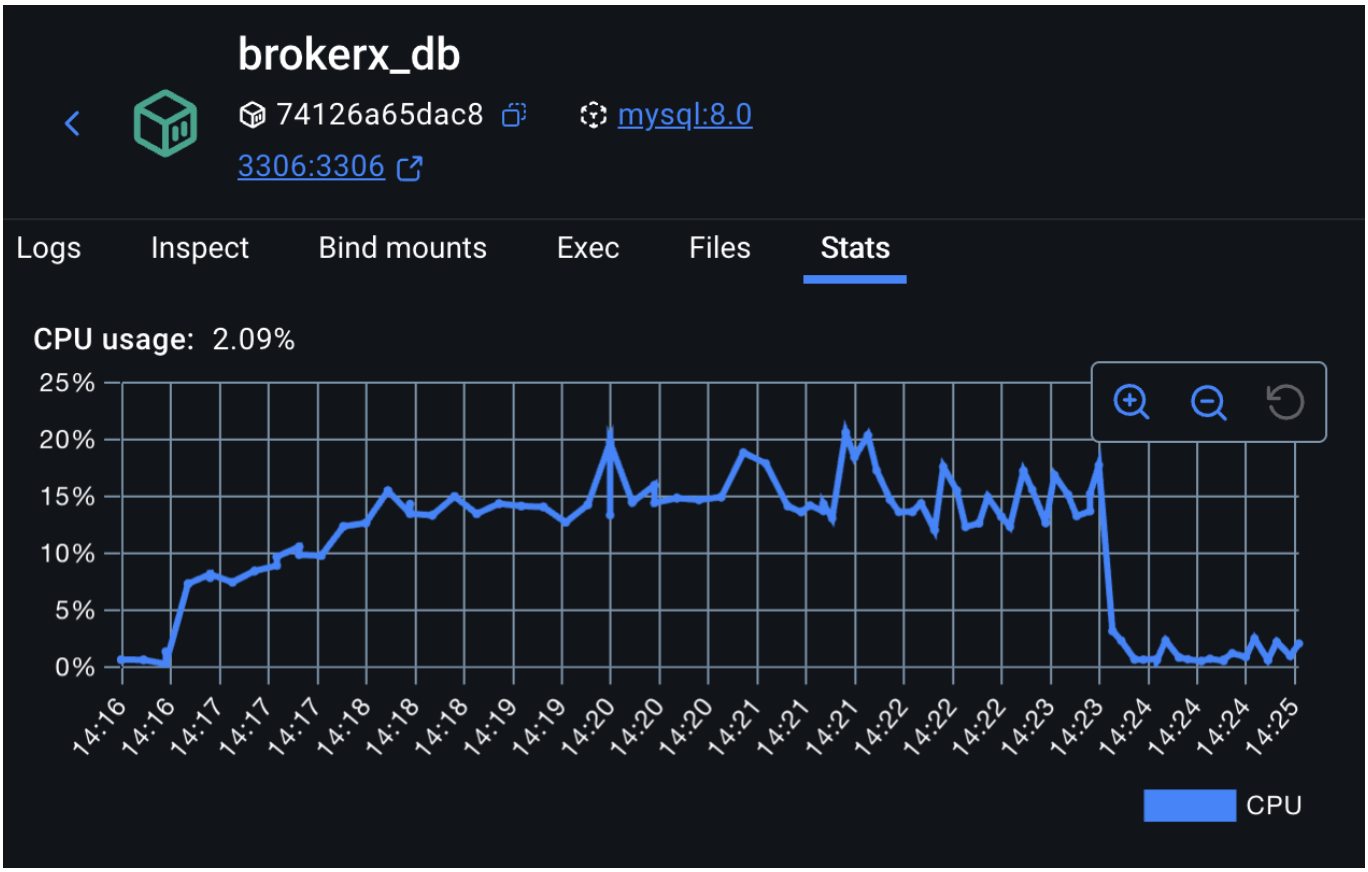


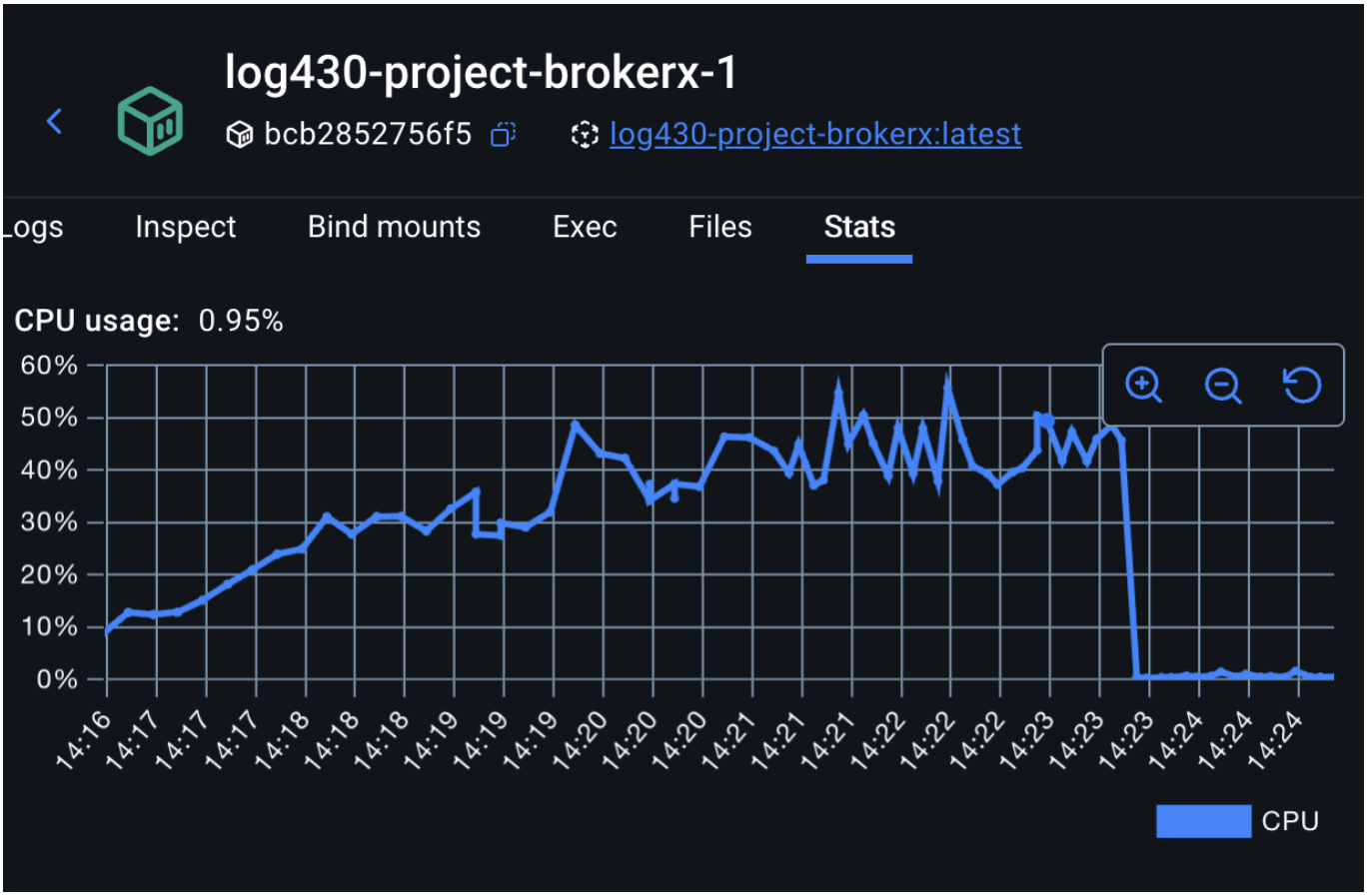


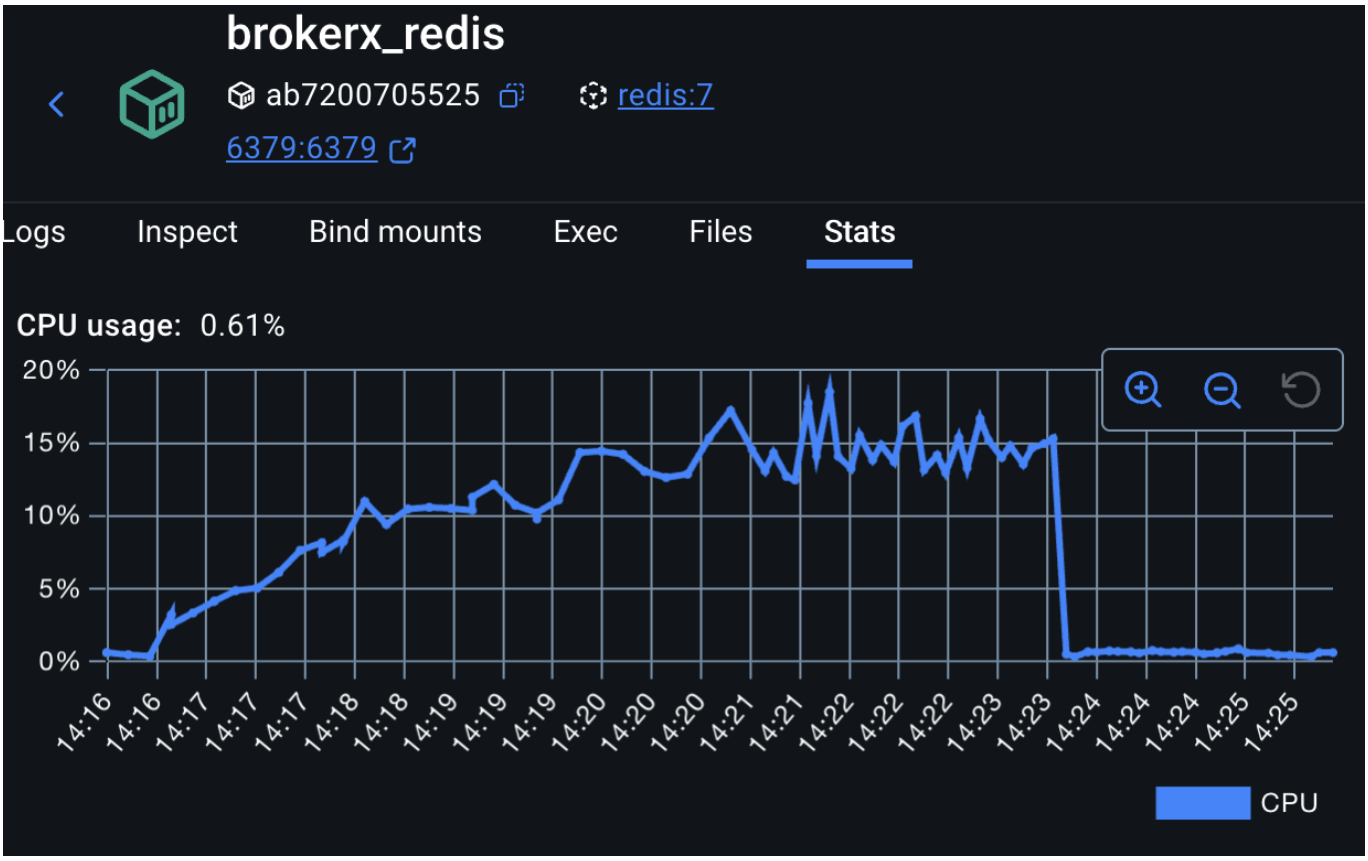
It seems db cpu usage decreased a little bit. Overall performance seems more stable.

Performance test result after implementing Redis order persistence queue and execution record persistence queue (7 minutes, 500 users, 5 users/second) :




brokerx_db
74126a65dac8
mysql:8.0
3306:3306
Logs
Inspect
Bind mounts
Exec
Files
Stats
CPU usage: 2.09%
25%
20%
15%
10%
5%
0%
14:16 14:16 14:17 14:17 14:17 14:18 14:18 14:18 14:19 14:19 14:20 14:20 14:20 14:21 14:21 14:21 14:22 14:22 14:22 14:23 14:23 14:24 14:24 14:24 14:25
CPU

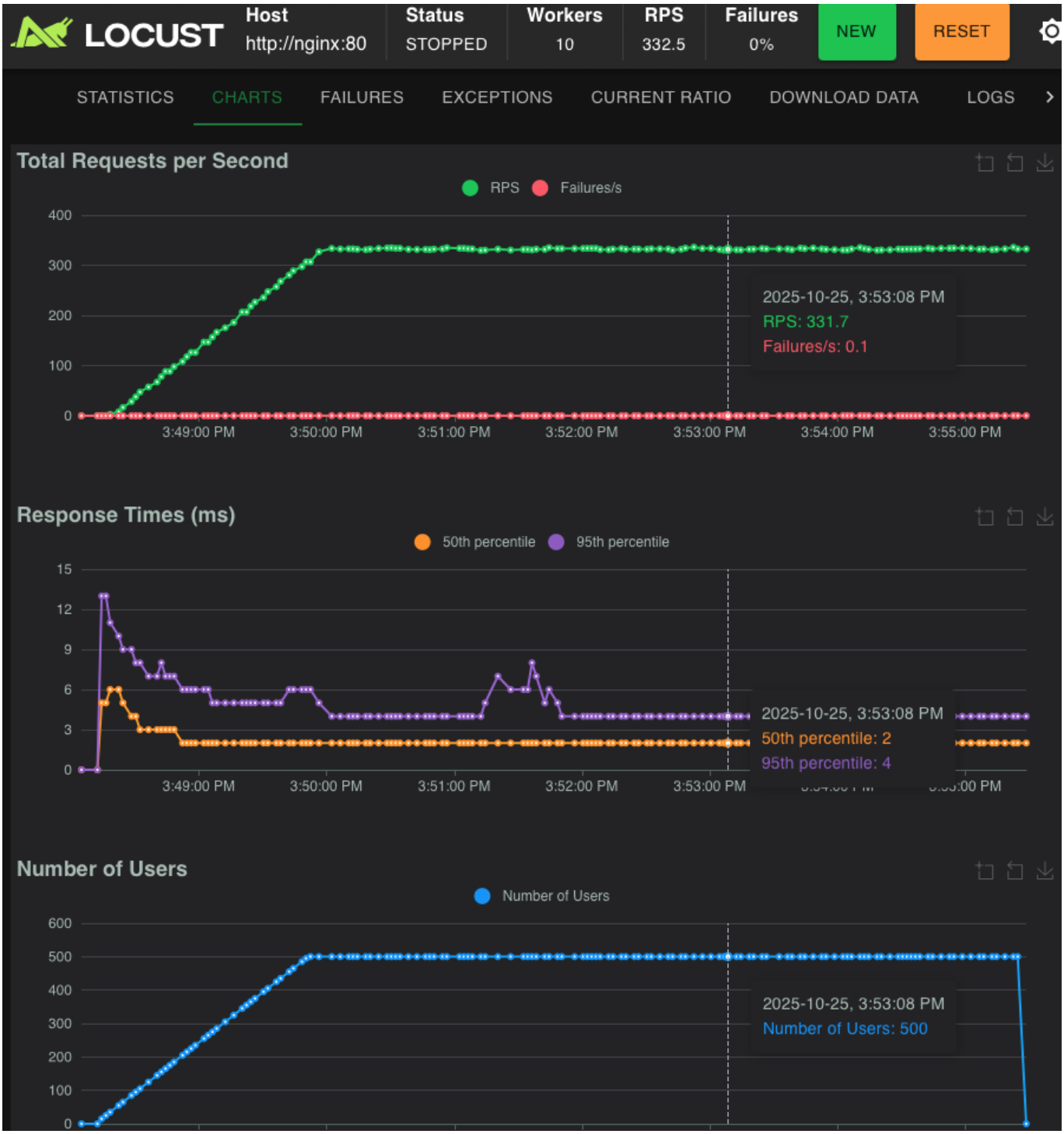

log430-project-brokerx-1
bcb2852756f5
log430-project-brokerx:latest
Logs
Inspect
Bind mounts
Exec
Files
Stats
CPU usage: 0.95%
60%
50%
40%
30%
20%
10%
0%
14:16 14:17 14:17 14:17 14:18 14:18 14:18 14:19 14:19 14:19 14:20 14:20 14:20 14:21 14:21 14:21 14:22 14:22 14:22 14:23 14:23 14:23 14:24 14:24 14:24
CPU



The goal of these improvements was to reduce the over reliance on the sql database and I think this goal was achieved. As we can see, with a stable load of requests (~300 RPS), the cpu load of each key component seems to stabilize (even though the number of stored orders is constantly increasing) and especially the database cpu load has been reduced significantly with plenty of room for higher load testing.

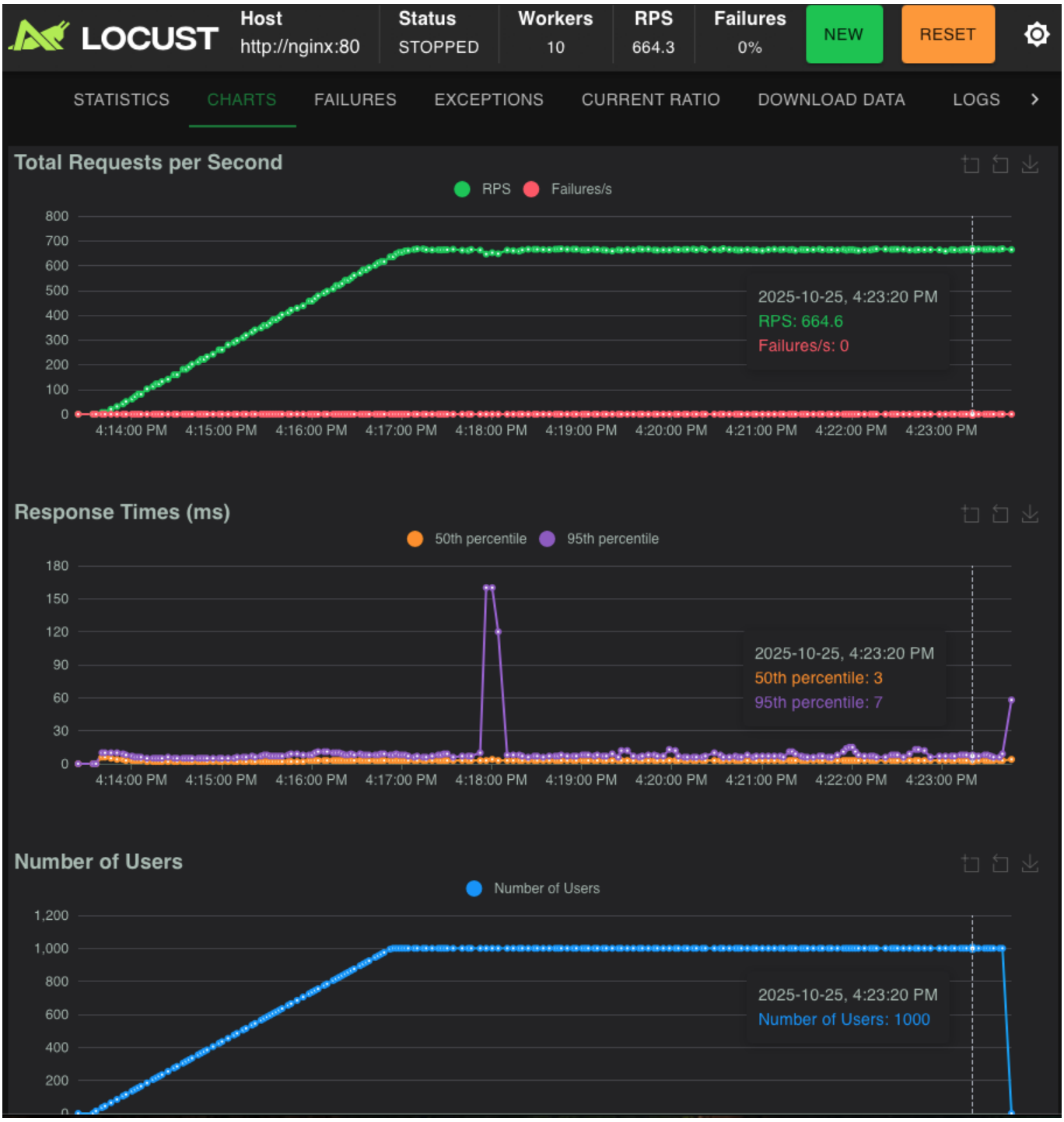
Added load balancing

Performance test result 2 instances (7 minutes, 500 users, 5 users/second)



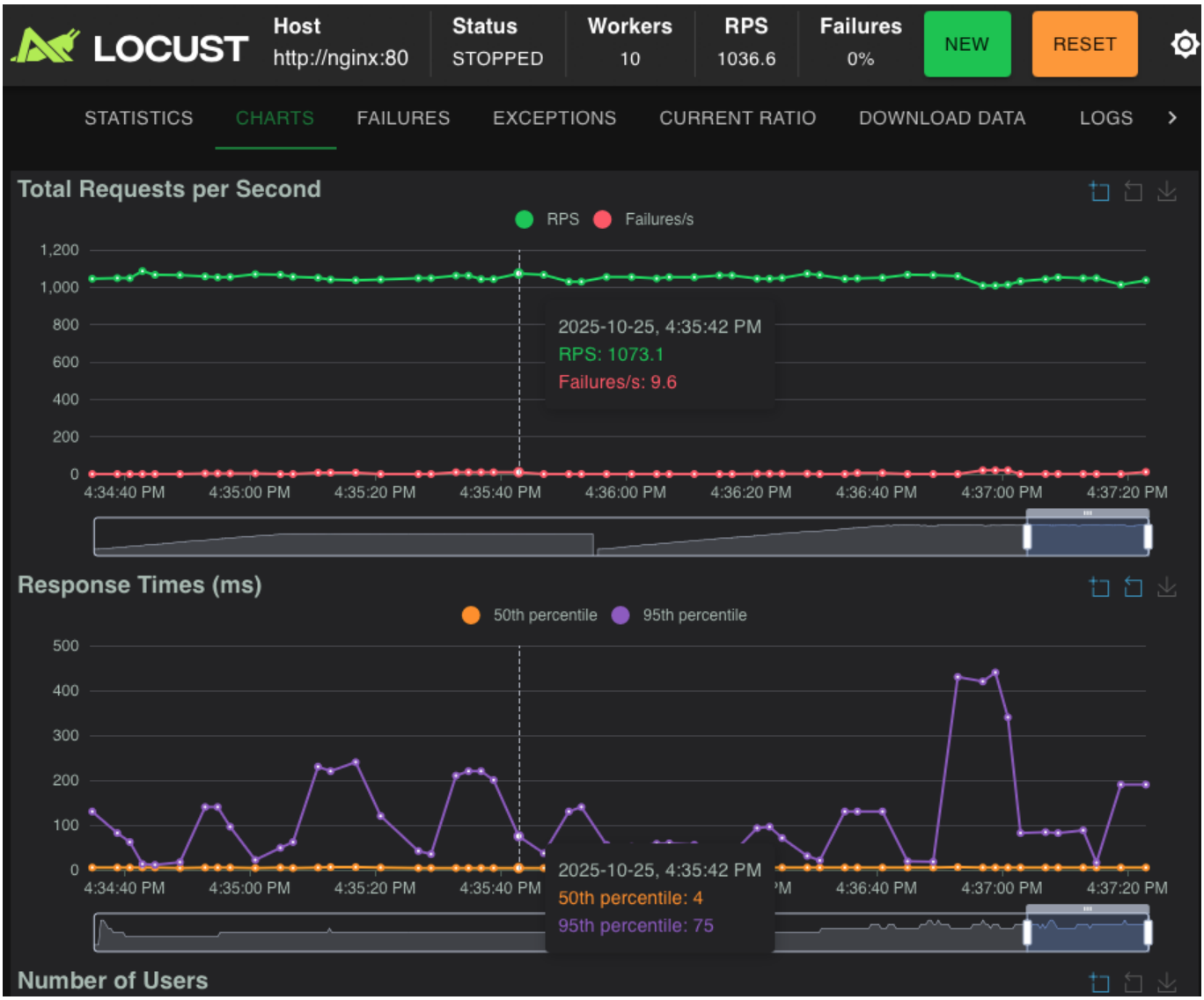
There seems to be no need for more than two instances at this level of request load. All metrics are very stable with each brokerx instance using around 10% cpu.

Performance test result 2 instances (10 minutes, 1000 users, 5 users/second)



VERY acceptable latency, cpu loads are very stable.

Performance test result 2 instances (10 minutes, 1600 users, 5 users/second)



With a request load of 1000+ RPS, the app is still comfortably responding within 250 ms with an average latency of 5 ms! The closest bottleneck still seems to be the sql database, meaning that future improvements should focus on increasing batch sizes when persisting to database and less reliance on database for order processing and matching.

We settle at 2 instances of the go api with the current monolith setup.

Phase 2 : Micro-services architecture BrokerX

We expect that transitionning to micro services should help distribute the load between the various services and therefore improve performance.

However, micro-services bring about a new challenges such as http client connection pooling and increased latency because of calls between the services.

Below are the results of the few performance tests that were performed after the transformation of BrokerX into RESTful API micro-services with an NGINX API Gateway.

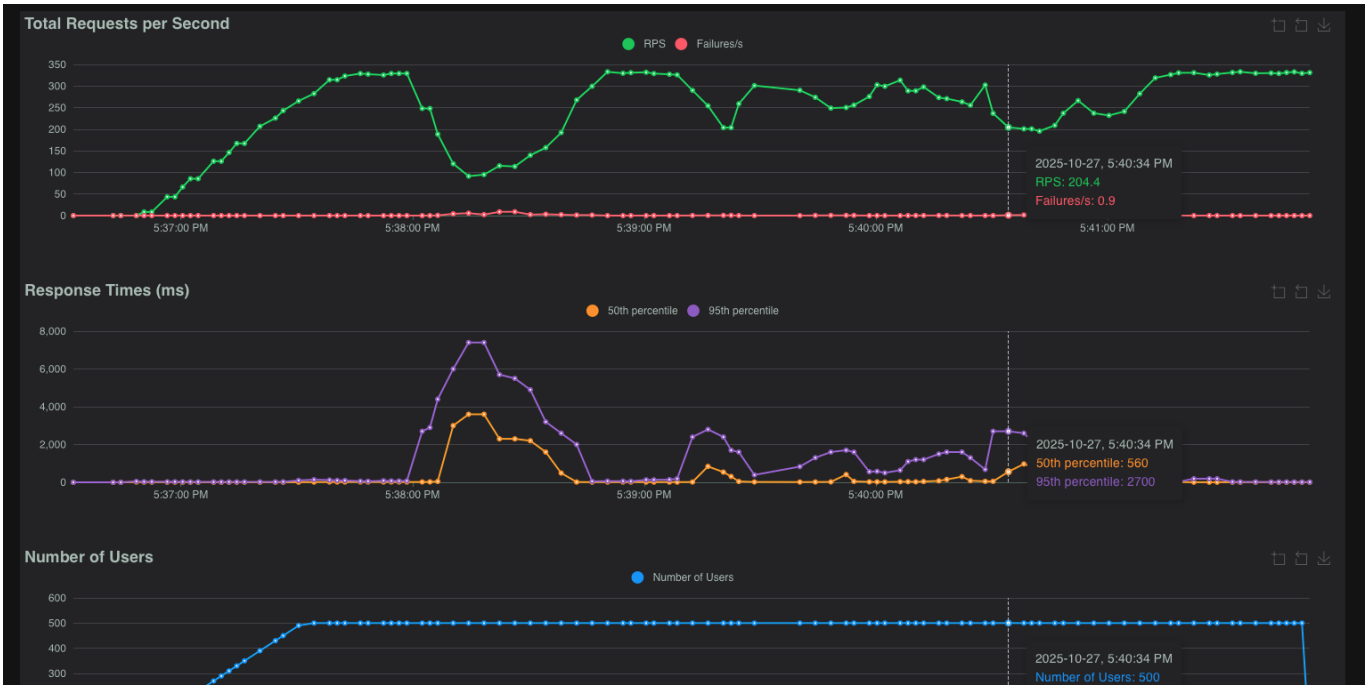
No load balancing

Run #1 (300 RPS)

First trying out performance tests on new microservice api gateway architecture, and the results are not great when reaching over 300 RPS. The order service becomes overwhelmed because the requests to market-service and portfolio-service dont work anymore due to nginx 512 worker_connections are not enough

Run #2 (300 RPS)

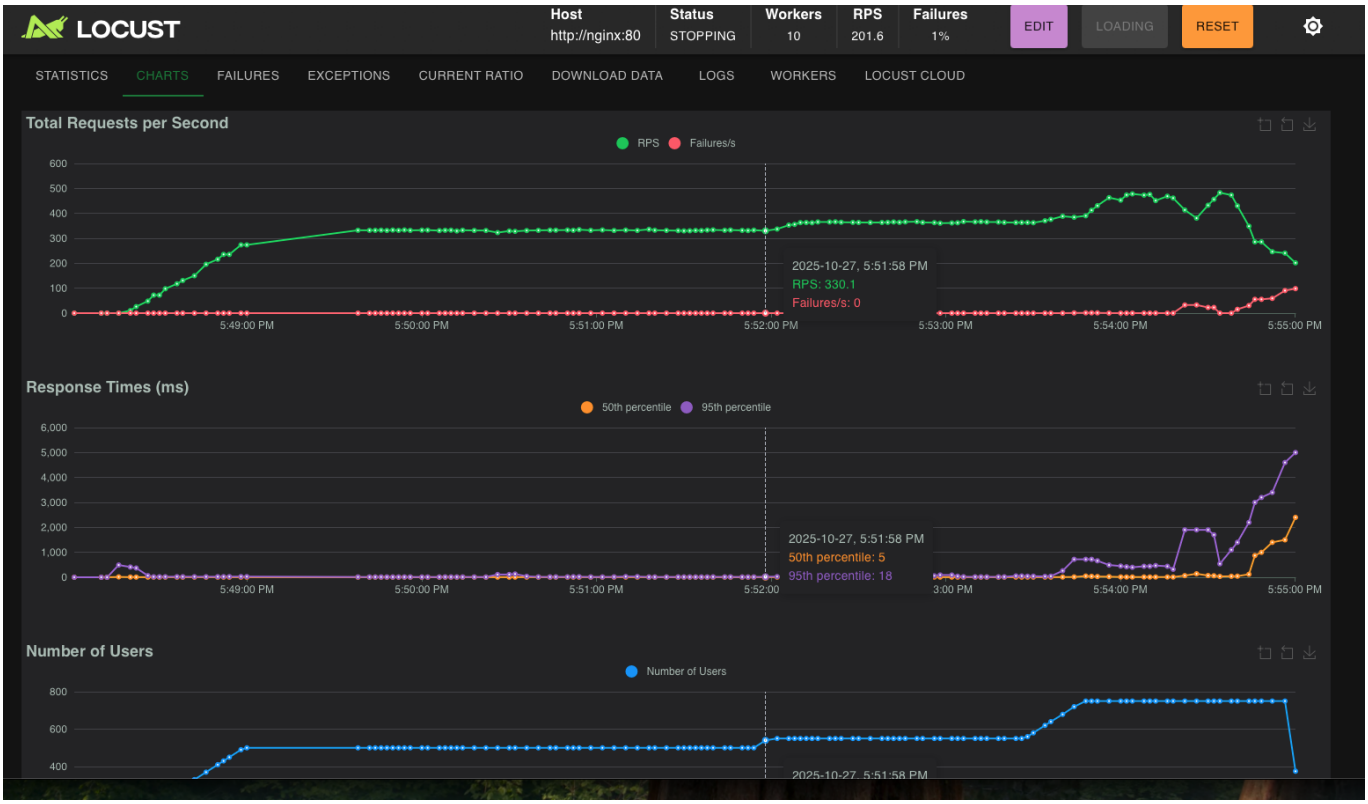
First improvement : increase nginx worker connections



Less errors but the delay is unacceptable

Run #3 (300 RPS)

Second improvement : allow each service to keep up to 100 idle tcp connections

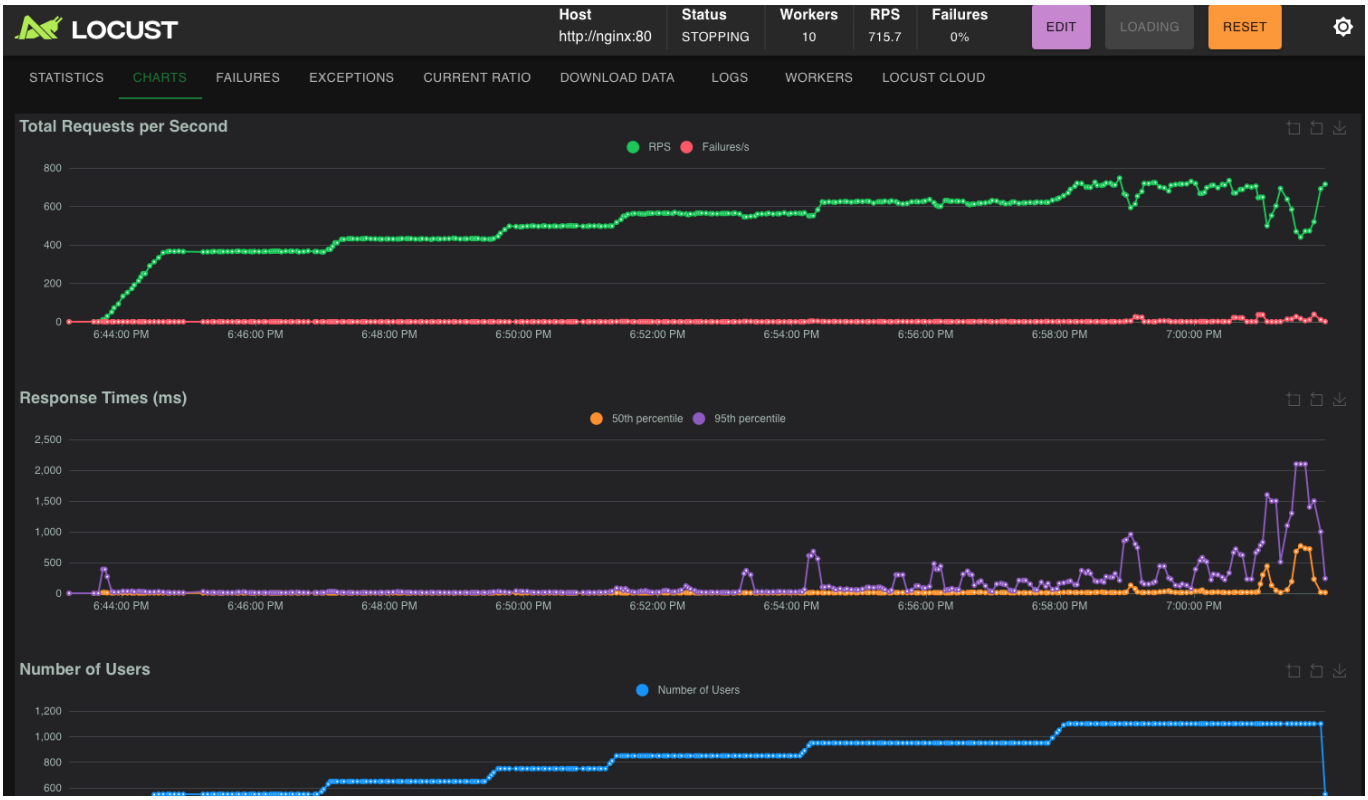


Improvement but failures and high latency start to appear again once reaching 400 RPS. The culprits are the following in the order service :

- level=error msg="matching failed for order #132124: error making request to matching service: Post "http://nginx/api/matching/": read tcp 172.18.0.7:38756->172.18.0.10:80: read: connection reset by peer"
- level=error msg="matching failed for order #133737: error making request to matching service: Post "http://nginx/api/matching/": context canceled"
- level=error msg="matching failed for order #132440: error making request to matching service: Post "http://nginx/api/matching/": http: server closed idle connection"

Run #4 (up to 600 RPS)

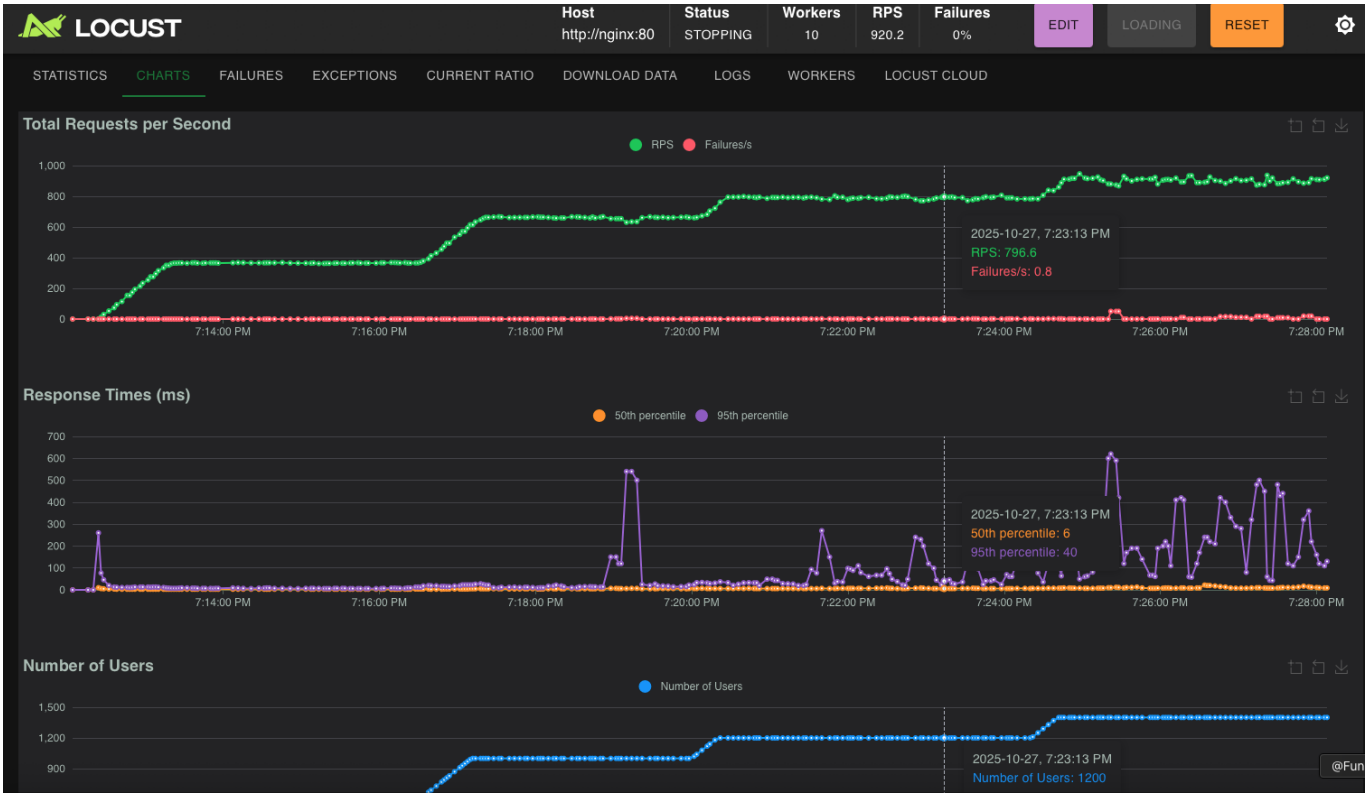
Third improvement : Add shared http client for order service



System gets unstable quickly after 600 RPS.

Run #5 (up to 900 RPS)

Fourth improvement : Make calls from one microservice to another internal calls that dont go through nginx.



System gets abit unstable when reaching around 800 RPS. Surprisingly high cpu usage on order service and matching service. They are the main service doing work but monolith brokerx never reached this high of a cpu load.

With load balancing

> Load balancing was not tested on the microservices architecture due to the lack of CPU resources available.

Phase 3 : Event-driven architecture BrokerX

To be done

Load Tests Conclusions

The load tests showed how BrokerX's architecture and code evolved to achieve the desired results. The initial monolithic version from phase 1 had a clear overreliance on the database and took care of too much static html rendering.

Progressive improvements were implemented, such as Redis caching and batch writes to the database, and this is where most of the performance was improved.

When transitionning to microservices, new problems arose, but were fixed to allow for 800+ RPS an average latency under 250 ms and an availability over 99%.